Policy Research 2024-19-04

2024 K-Innovation Partnership Program with Cambodia

Policy Consultation and Pilot Implementation on Developing of National R&D Management System and R&D Center of Excellence in Cambodia

SCIENCE AND TECHNOLOGY POLICY INSTITUTE

2024 K-Innovation Partnership Program with Cambodia

Title	Policy Consultation and Pilot Implementation on Developing of National R&D Management System and R&D Center of Excellence in Cambodia
Institute	Science and Technology Policy Institute (STEPI)
Partner country	The Kingdom of Cambodia
In cooperation with	Ministry of Industry, Science, Technology, and Innovation (MISTI)
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Executive Summary

The main objective of the K-Innovation ODA Programme is to build and enhance the STI policy capacities of each partner country, ranging from design and implementation, to evaluation and feedback. The K-Innovation ODA Programme's wealth of professional and academic experience has enabled the Kingdom of Cambodia (Cambodia) project to produce important national strategies and technology roadmaps.

This is the second year of the Phase 3 project of the K-Innovation Programme with the Ministry of Industry, Science, Technology & Innovation (MISTI) of the Royal Government of Cambodia (RGC). The Phase 1 project, conducted from 2018 to 2019, established Cambodia's first national plan on science and technology and resulted in the launch of "STI Policy 2020-2030" as the first official STI Master Plan of the RGC. Subsequently, the Phase 2 project (2020 to 2022) developed six national technology roadmaps (Education, Health, Agriculture, Digital, Energy, and Tourism) related to the implementation of the National STI Master Plan. Despite the COVID-19 restrictions, the project was managed successfully and conducted effectively thanks to the joint efforts of STEPI and MISTI. The Phase 3 project, which will be carried out from 2023 to 2025, is focused on the establishment of a national R&D management system in Cambodia.

The consultation process for the project in 2024 consisted of the following main activities:

As a first step, STEPI and MISTI held a kick-off meeting on 13 February 2024 and agreed on the key activities and schedule of the 2024 programme. Then, STEPI participated in Cambodia's second "National STI Day" event held on 23-28 March 2024. During this event, STEPI and MISTI held a Tech Show to share the findings of the needs assessment of the first year activities of the K-Innovation Cambodia project and to demonstrate the National Science and Technology Information Service (NTIS) of Korea. The assessment identified the most critical requirements of Cambodian researchers regarding national R&D programs. Key needs included access to project announcement, researcher profiles, and research outcome data, emphasizing the necessity of establishing a comprehensive platform. Furthermore, NTIS was introduced as a platform that provides integrated information on national R&D activities for researchers, students and policymakers.

The second step consisted of a stakeholder workshop, as a side event of ASEAN-COSTI 85 held on June 6th, 2024, in Siem Reap Cambodia. The event featured the presentations of the initial system design draft and the collection of feedback from key stakeholders. The first session contains background and needs assessment results for the Cambodia's National R&D Information Management System (CTIS). The second session topic was a concept and development plan for the CTIS. As a panel discussion, timelines of implementation, anticipated benefits and R&D budget were discussed.

As a third step, a service contract was signed between STEPI and MISTI. The contract focuses on assessing the feasibility of building a CTIS and includes reviewing relevant legal and institutional support framework on R&D and conducting a needs assessment.

Finally, overall project efficiency and effectiveness were carefully managed throughout the project process. STEPI and MSITI have worked together very closely to implement the project's key activities.

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Research Fellow, STEPI

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Honorary Research Fellow, STEPI

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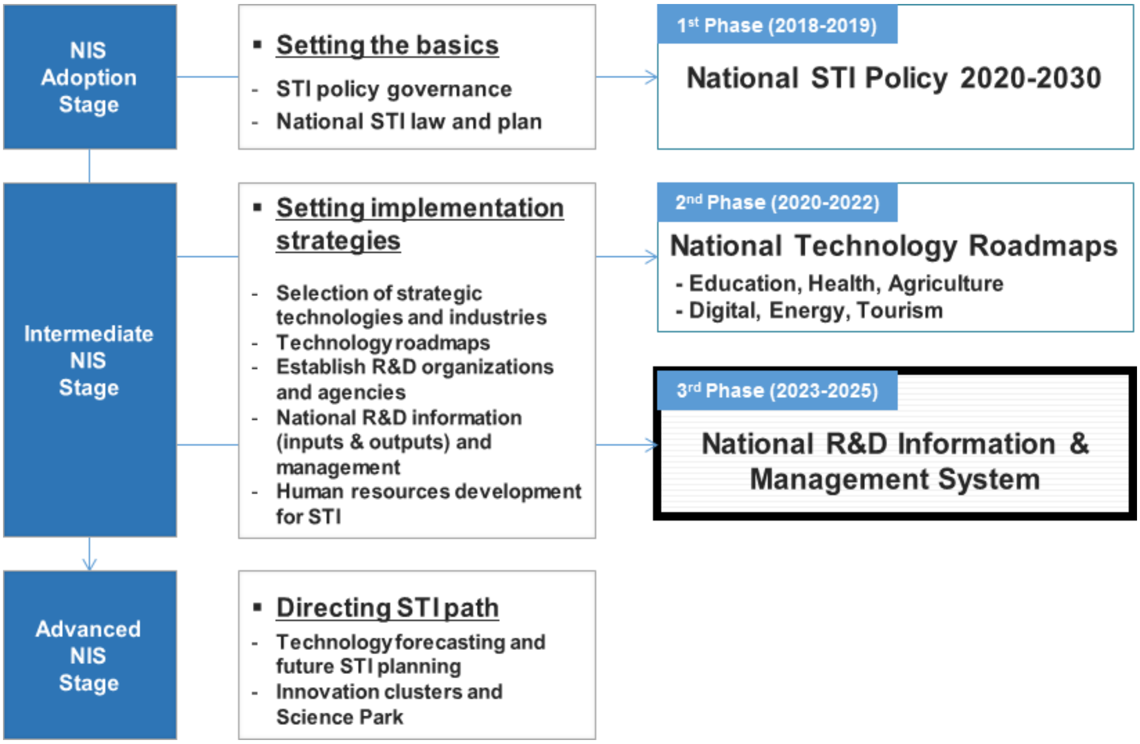
Chapter 1. Project Overview

1. Introduction

This K-Innovation Partnership Programme with Cambodia is currently in the second year of its third phase (2023-2025), titled “Project Consultation and Pilot Implementation for the Development of the National R&D Management System and the R&D Centre of Excellence in Cambodia”. The partner organisation is the Ministry of Industry, Science, Technology & Innovation (MISTI).

During the first phase of the K-Innovation Programme (2018 to 2019), STEPI provided policy consulting on STI policy governance and National STI law and supported the Cambodian government in developing its first National STI Master Plan and National STI Policy 2030-2030. In the second phase (2020 to 2022), the STEPI team collaborated with Cambodian stakeholders to develop a national technology roadmap (TRMs) for each of six major technology areas, namely, education, health, agriculture, digital, energy, and tourism. Three technology roadmaps (education, health, and agriculture) were adopted by Cambodia’s National Council on Science, Technology & Innovation in July 2021, while the other three technology roadmaps were launched on Cambodia’s first “National STI Day” in March 2023. As part of the second phase of the K-Innovation Programme with Cambodia, equivalent to an intermediate “National Innovation System (NIS)” stage, the activities included selecting strategic technologies and industries, developing technology roadmaps, and improving capacity building for STI policy. The third phase of the programme is also a part of the intermediate NIS stage, providing policy consultation and supporting capacity building for the development of the National R&D Management System for the Cambodian government, as shown in Figure 1-1 below.

[Figure 1-1] Overview of the K-Innovation Partnership Programme Phases 1-3



Source: Kim et al. (2023).

2. Objectives

The overall objectives of the third phase of K-Innovation Partnership Programme with Cambodia are to provide policy consulting aimed at advancing the National R&D Management System in Cambodia in order that its STI Master Plan and TRMs can be implemented fully, and at building the skills and capacities of the researchers and the related policymakers. In 2024, the programme focused on designing the conceptual structure and detailed components for the national R&D management system. Based on the environmental survey and analysis of Cambodia's national R&D management system conducted in the first year (2023), along with the sharing of Korea's experiences in building a national R&D management system, the second year (2024) focused on providing advisory services for the design of Cambodia's National R&D Information System (CTIS). Additionally, consultations were provided on national science and technology standard classification systems, referencing advanced practices from Asian countries such as Korea, China, and Japan.

For the next year, 2025, STEPI's research team will provide an implementation plan of establishing CTIS and support and assist MISTI's capacity building in relation to the national R&D management system, which will be tailored to Cambodia's needs and its national innovation system.

3. Project Framework

The framework of the K-Innovation Partnership Programme has two major components: policy consulting and capacity building. Together with invited experts from the Korea Institute of Science and Technology Information (KISTI) and National Institute of Green Technology (NIGT), STEPI provided policy consulting mainly covered providing advisory services for the conceptual design of Cambodia's national R&D information system. In consultation with STEPI, MISTI analysed the key functions and data required to be monitored based on the last years' survey results holding a series of consultation meetings with local stakeholders to understand the status of, and identify demand for, the national R&D management system in Cambodia. Additionally, a series of consultations was provided on national science and technology standard classification systems, referencing advanced practices from Asian countries such as Korea, China, and Japan.

4. Project Team

■ Korean Research Team

The STEPI research team was composed of three researchers, a PM, an advisor, and a project coordinator. To ensure the continuity of the multi-year development aid project, Dr. PARK Dong Un took the PM position after participating in the 2022 programme with Cambodia. Dr. LEE Jungwon, an honorary research fellow of STEPI and head of the first and second phases of the programme with Cambodia, participated as an advisor in the 2024 programme.

Two KISTI experts with expertise in the national R&D management system were invited to advice on the creation of CTIS concept design with key functions and data that is required to be collected and monitored.

[Table 1-1] Korean research team

Category	Institute	Position	Name	Major role
Internal expert	STEPI	Research Fellow	Dr. PARK Dong Un	Overall management of the 2024 Programme with Cambodia, research and surveys)
	STEPI	Honorary Research Fellow	Dr. LEE Jungwon	Advisor
	STEPI	Senior Researcher	Ms. CHUNG Hailey Hyorim	Project coordination, research and surveys
External expert	KISTI	Senior Researcher	Dr. PARK Kuen Hee	Policy consulting on NTIS
	KISTI	Senior Researcher	Dr. YU Eunji	Policy consulting on NTIS

■ Cambodian Research Team

The Cambodian research team was headed by Dr. HUL Seingheng, the Under-Secretary of MISTI. Under his leadership for the third phase of the programme, Dr. SRUN Pagnarith served as a project coordinator for the Cambodian team as well as a counterpart for the STEPI team. Three other officials from MISTI who are responsible for STI policy and STI data management were also members of the team involved in the 2024 Programme and the national R&D management system.

[Table 1-2] Cambodian research team

Category	Organization	Position	Name	Major role
Cambodian research team	Ministry of Industry, Science, Technology and Innovation (MISTI)	Under-Secretary	Dr. HUL Seingheng	Overall project management on the Cambodian side
		Deputy Director General of the STI Policy Department	Dr. SRUN Pagnarith	Project coordinator, research and surveys
		Director-General, General Department	Dr. TRY Sophal	Research and surveys
		Deputy Director of the STI Data Management Department, GDSTI	Dr. CHEN Sovann	Research and surveys
		Deputy Director	Dr. SENG Touch	Research and surveys

5. Project Schedule

The overall schedule of 2024 programme is as follows:

[Table 1-3] Project Implementation Schedule

Task/Month	1	2	3	4	5	6	7	8	9	10	11	12
Planning	■	■										
Revision of plan			■									
Kick-off workshop, onsite interviews			■									
Stakeholder Workshop In Sime Reap						■						
Mid-term Evaluation (STEPI internal)							■					
Signing of local service contracts									■	■		
Final evaluation (STEPI internal)											■	■
Final report												■

6. Project Outputs

STEPI provided the following deliverables:

- Deliverable 1: Kick-off workshop presentation materials.
- Deliverable 2: Materials of a stakeholder workshop as a side event of ASEAN-COSTI 85 in Cambodia
- Deliverable 3: Cambodia's National R&D Information Management System (CTIS) Concept and Development Plan of Actions

MISTI provided the following deliverables:

- Deliverable 1: Report of the status of legal and regulatory supporting framework and risk assessment in Cambodia
- Deliverable 2: Khmer version report of the feasibility study on the establishment of the Cambodia National R&D Management System and key stakeholder hearing

CHAPTER 2

Main Activities

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Research Fellow, STEPI

Ms. Hailey Hyorim Chung
Researcher, STEPI

Chapter 2. Main Activities

1. Main Activities

1.1 Programme adjustment and operation meetings (online)

13 February 2024: A kick-off meeting was held between the STEPI and the MISTI to discuss and outline the key activities and the detailed schedule for the 2024 programme. The meeting provided an opportunity for both organizations to align their objectives and expectations, ensuring a clear understanding of the tasks, timelines, and responsibilities associated with the programme. Additionally, this meeting served as a platform to address any initial questions or concerns and to establish a collaborative framework for the successful execution of the planned activities throughout the year.

1.2 The Tech Show and Kick-Off Workshop during the Cambodia's National STI Day

24 March 2024: The workshop included a comprehensive presentation of the findings from the National R&D Management System demand survey, followed by a detailed demonstration of the NTIS platform. These activities were complemented by in-depth discussions on the system's relevance to Cambodia's R&D landscape and its potential applications in addressing the country's pressing R&D needs effectively. The session provided participants with a clear understanding of how such a platform could be utilized to support the development of Cambodia's research ecosystem.

- The demand survey, which was conducted as part of the first-year K-Innovation Cambodia initiative, successfully identified the key needs of Cambodian researchers engaged in national R&D programs. The findings highlighted the urgent necessity for establishing an integrated platform that provides comprehensive information on project announcements, researcher profiles, and research outputs. This platform would serve as a critical tool for enhancing the efficiency and visibility of R&D activities in Cambodia. (Presented by Dr. Srun, Deputy Director, MISTI, Cambodia)
- Korea’s National Science & Technology Information Service (NTIS) serves as a powerful, integrated platform offering detailed and up-to-date information on national R&D activities. NTIS is designed for use by researchers, students, and policymakers, providing essential resources to support the R&D process. (Presented by Dr. Dong Un Park, Research Fellow, STEPI; Demonstrated by Ms. Hailey Hyorim Chung, Researcher, STEPI)
- NTIS facilitates easy access to crucial statistics required for effective policy development. It includes comprehensive data on R&D personnel, budget trends, and performance outcomes, all of which are essential for monitoring and shaping national R&D strategies.
- Research Collaboration: NTIS allows researchers to search for potential collaborators by accessing detailed profiles, which include information on affiliations, research histories, and accomplishments. The platform also enables users to initiate collaboration requests directly, making it easier to connect with other researchers in their field.
- Research Information: Users can search for specific topics, such as "fine dust," and retrieve a wealth of related information, including ongoing research projects, researcher profiles, R&D institutions, academic papers, patents, reports, and international projects. This comprehensive resource streamlines the research process and provides valuable insights for researchers across different disciplines.

- Project Insights: The "Research Projects" tab on NTIS provides detailed information on various research initiatives, including project budgets, researcher composition, funding sources, project types, and summaries. This feature allows users to gain a deeper understanding of the scope and impact of different research projects.
- Shared Resources: NTIS also facilitates the identification and reservation of shared research facilities and expensive equipment. The platform provides users with information on available equipment models, specifications, locations, and the contact details of the institutions that own these resources, promoting efficient resource sharing.
- Research Outputs: Research outputs are organized and categorized by keywords, national strategic technologies, and technological trends. Users can access visualized data on these outputs, which helps in understanding the broader research landscape. Additionally, the platform allows users to identify the funding ministries supporting specific R&D areas.
- Convenience in Project Announcements: Through NTIS's integrated R&D announcement service, users can view the latest national R&D opportunities and opt to receive email updates for newly announced projects. This ensures that stakeholders are kept informed of the latest developments in the R&D sector.
- Insights for Policymakers: NTIS offers a wealth of critical science and technology statistics, such as total R&D expenditures, stages of R&D activities, personnel data, technology trade, and innovation metrics. This data is invaluable for policymakers in formulating evidence-based science and technology policies. (Presented by Dr. Jung Won Lee)
- NTIS allows users to access data specific to different sectors, such as energy, space, and biotechnology. This enables targeted research and policy analysis, making it easier for users to focus on the data most relevant to their interests or areas of work.
- 27 March 2024: The kick-off workshop was organized to facilitate the second-year implementation of the K-Innovation Partnership Programs in Cambodia (2023–2025). This workshop aimed to provide an overview of the upcoming initiatives and set the stage for the continued collaboration between the participating institutions.

- As part of the second-year initiatives under the Cambodia International Technology Innovation Cooperation Project, a series of plans are being undertaken to align with Cambodia's ICT strategy and to establish an effective Research and Development (R&D) Management System. These initiatives are aimed at improving the country's R&D infrastructure and promoting a more robust and integrated approach to technological innovation.
- A structured framework and detailed execution plan will be developed to guide the establishment of the R&D management system. This plan will be designed to ensure alignment with Cambodia's national ICT strategy and also leveraging existing resources and information to optimize the system's effectiveness.
- Key tasks for the second year include defining the necessary information and services, standardizing processes, designing system prototypes, determining the scope of the system, and estimating the budget for implementation. These tasks are critical to ensuring that the R&D management system meets the needs of Cambodian researchers and policymakers.
- Based on the findings of last year's survey on local conditions, the system design—including the required functionalities and data collection plans—will be finalized. Budget estimates are scheduled to be completed by May to ensure that the project stays on track and within budget.
- MISTI will collaborate closely with the Korean KISTI NTIS team by providing a detailed inventory of the data stored in its data centers. This inventory will include information on hardware, software, and other infrastructure, which will be essential for accurately estimating the budget for the system's development.
- Clearly articulating MISTI's requirements will be key to ensuring that the system is tailored to local needs, rather than adopting a generic solution. This approach will foster greater usability and relevance, ensuring that the system can effectively address Cambodia's specific R&D challenges and support its research community.

1.3 A Stakeholder Workshop as a Side Event of ASEAN-COSTI 85 in Cambodia

- 6 June 2024: STEPI held a stakeholder workshop, as a side event of ASEAN-COSTI 85 held on June 6 2024, in Siem Reap Cambodia. The event featured the presentations of the initial system design draft and the collection of feedback from key stakeholders. The first session contains background and needs assessment results for the Cambodia's National R&D Information Management System (CTIS). The second session topic was a concept and development plan for the CTIS. As a panel discussion, timelines of implementation, anticipated benefits and R&D budget were discussed.
- The Cambodian government recognizes science, technology, and innovation (STI) as key drivers of sustainable socio-economic development, with research and development (R&D) being a cornerstone for fostering a knowledge-based and innovation-driven economy.
- Key findings from the needs assessment revealed that the current R&D management system is composed of 61.9% applied research and 41.79% basic research. Collaboration is primarily facilitated through networking and memorandum of understanding (MOU) agreements, and most projects rely on funding from development partners (63%).
- However, the existing system faces significant limitations, including insufficient tools for securing research funding, managing research activities and reports, and tracking progress and milestones. To address these gaps, the system should aim to streamline project planning and resource allocation, enhance collaboration and communication among researchers, and emphasize the importance of data visualization and analysis functionalities to promote a vibrant research ecosystem.
- Leadership at the institutional level is critical for the successful implementation of a national R&D management system. Key functionalities required include the establishment of a centralized research database, systems for submitting project proposals, resource management tools, and databases for publications, citations, and

researcher expertise.

- Concept and Development Plan for the Cambodian Research and Development Information Management System (Presentation by Dr. Siev Sokly, Deputy Director General, MISTI Cambodia)
- The envisioned NTIS aims to systematically manage and utilize national research and scientific data.
- Technical Benefits: The proposed system is expected to enhance research productivity and generate significant economic and industrial spillover effects. It will also support the reliable utilization of national R&D outcomes.
- Economic Benefits: By supporting research activities and enabling effective use of information, the system is anticipated to increase research productivity and amplify economic impacts.
- Development Plan: A detailed roadmap includes defining information and services, standardization plans, prototype development, and budget estimation.
- Information and Service Definition: Conduct surveys to assess the current status of R&D information management in Cambodia, analyze the needs of the scientific community, and evaluate national priorities.
- System Objectives: Gather internal feedback from MISTI to define the goals of the R&D information management system.
- Roadmap and Budget: Develop a comprehensive roadmap and calculate the budget required for system implementation.
- Panel Discussion
- (Timeliness of Implementation) The Cambodian government's plan to establish a national research foundation as a funding agency aligns well with the introduction of a digital national R&D information system. This system will enable efficient monitoring, investigation, analysis, and evaluation of national R&D programs, making it highly timely and relevant.

- (Implementation Approach) Engaging a wide range of stakeholders is essential. Policymakers should invite representatives from ministries beyond MISTI, as well as researchers from universities and institutes, to hear diverse perspectives. Addressing data security concerns by proactively incorporating stakeholder feedback during system development is also critical.
- (Anticipated Benefits) Korea's NTIS serves as a successful example. Over 20 years, it consolidated dispersed information on national R&D projects across various ministries into a single platform. This integration helped prevent redundant investments, optimized budget use, and provided foundational data for researchers, such as preventing duplicate proposals. With such a system, the Korean government was able to confidently expand its national R&D budget.
- (R&D Budget Considerations) Cambodia's STI Master Plan 2030 outlines a goal of increasing R&D investment to 1% of GDP (0.5% public and 0.5% private). Korea's historical experience offers valuable insights. In its early stages, public investment accounted for 70–80% of R&D funding, with private sector contributions increasing significantly only after proactive government support for private R&D investments. Cambodia should adopt a similar approach, initially prioritizing government-led investments while introducing policies to stimulate private sector participation.

1.4 A Service Contract Signed Between STEPI and MISTI

- 7 October 2024: A service contract was officially signed between STEPI (Science and Technology Policy Institute) and MISTI (Ministry of Industry, Science, Technology, and Innovation). The purpose of this contract is to assess the feasibility of developing a CTIS. This assessment will include a detailed review of the relevant legal and institutional frameworks supporting research and development (R&D) in Cambodia, as well as conducting an in-depth needs assessment. The full scope and findings of this process are discussed in detail in Chapter 3 of this report.

- The main objective of the contract is to prepare a comprehensive feasibility report for the establishment of Cambodia's National R&D Management System. To achieve this, the contract involves performing a thorough review of the existing legal and institutional frameworks that support R&D activities, as well as conducting a detailed needs assessment to identify the requirements for a robust and effective national system. The insights from this assessment will guide the development of the proposed system.
- Key activities outlined in the contract include conducting an in-depth review of the legal and institutional frameworks that are relevant to the establishment of the system. This review will be carried out through a series of expert interviews, discussions, and meetings with stakeholders involved in R&D activities. Additionally, the feasibility report will be translated from English into Khmer to ensure its accessibility for local stakeholders, and a stakeholder consultation workshop will be organized to gather feedback and refine the proposal.
- The contract will span from October 7, 2024, to December 15, 2024. During this period, the deliverables, including the feasibility report in English and its Khmer translation, will be used as critical resources for conducting stakeholder consultations in Cambodia. These consultations will provide an opportunity for local stakeholders to contribute their insights and perspectives on the proposed system.
- Furthermore, the findings and recommendations outlined in the report are intended to directly inform the policymaking processes of MISTI or the National Council for Science, Technology, and Innovation (NCSTI). The integration of these findings into national policies will play a pivotal role in shaping the future of Cambodia's R&D landscape and ensuring that the country's scientific and technological development is effectively supported by appropriate institutional frameworks.

CHAPTER 3

Cambodia's Legal and Regulatory Supporting Framework on R&D

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Chapter 3. Cambodia's Legal and Regulatory Supporting Framework on R&D

1. Introduction

The Royal Government of Cambodia has set out to realize a vision to become a higher middle-income country by 2030 (Vision 2030) and a high-income country by 2050 (Vision 2050). To realize this ambitious vision, the government has been looking into emboldening the structural transformation, specifically transitioning from a labor-intensive economy to a knowledge-based economy, which highlights the importance of the role of Research and Development (R&D) underpinning the development of human capital, identify societal problems and tackle them, develop innovative methods and products, leverage national innovation capabilities of the country, all of which leading to create a robust, inclusive, and sustainable socio-economic development. To this end, the government of Cambodia, through various ministries and government agencies and institutions, has embedded R&D directly and indirectly into their respective policies and regulatory frameworks such as agriculture, energy, health, and industries.

This section aims to demonstrate the government of Cambodia's efforts in enhancing R&D in the national policies to leverage economic development and tackle the country's challenges and ultimately to realize the 2050 vision and its inclusive and sustainable development goal.

2. The National Core Policies

To achieve the Sustainable Development Goals and realize the 2030 and 2050 visions, the Royal Government of Cambodia has always disseminated its core policy flagship at the commencement of the new legislature of the national assembly. These policies cover all prioritized sectors and outline the policy agenda and targets that the government aims to achieve within its mandate.

2.1 Pentagonal Strategy Phase I

Cambodia's *Pentagonal Strategy Phase I* (2023-2028) for Growth, Employment, Equity, and Sustainability: Building the Foundation Towards Realizing the Cambodia Vision 200, which developed on the achievements of the Triangular Strategy and the Four-phase of Rectangular Strategy in the previous 25 years, a flagship policy of the Royal Government of Cambodia of the Seventh Legislature of the National Assembly, sets the groundwork for achieving Cambodia’s Vision 2050, emphasizes multiple pillars for national growth to ensure that in the next 25 years (1) Cambodia becomes a vibrant society that sustains everlasting peace, political stability and strong public order, (2) Cambodia becomes a high-income and resilient economy that is efficient, competitive, open, inclusive and sustainable, (3) Cambodian people are highly knowledgeable and have at least one skill in life with perseverance, entrepreneurial spirit, innovation and high morality, (4) Cambodian people live in dignity and happiness, and (5) Cambodia becomes a country that enjoys harmony, resilience, and inclusivity of physical and natural environment and has a good balance between development and environmental conservation. The strategy adopts five keys priority, namely People, Road, Water, Electricity, and Technology, in the consequential priority. Technology has put strong emphasis on digital innovation, scientific research, and skills development as critical to advancing the economy and modernizing society. To achieve the set forth objectives, the Pentagonal Strategy has rolled out five strategic pentagons outlined as Pentagon 1- Human Capital Development, Pentagon 2- Economic Diversification and Competitiveness Enhancement, Pentagon 3- Development of Private Sector and Employment, Pentagon 4- Resilient, Sustainable and Inclusive Development and Pentagon 5- Development of Digital Economy and Society. The technology and R&D goals within this framework aim to bolster digital infrastructure, promote a knowledge-based economy, and encourage digital innovation systems. Specific objectives include developing a digital government, encouraging digital business, and fostering an environment conducive to digital innovation. Moreover, the government aims to create new economic growth sectors and foster technical and vocational skills aligned with the demands of modern industry, further underpinning the importance of human capital development as foundational to R&D growth. The Pentagonal Strategy has also defined mechanisms to promote research such as Under the pentagon 1, the strategy to strengthen the governance of higher education institutions, promoting R&D and innovation, and promoting public private partnership. Under pentagon 4, the strategy seeks to promote scientific research in the agricultural sector, and under pentagon 5, the strategy sets to strengthen digital innovation capacity through development of human resources and talents in science, technology and innovation, as well as improvements of R&D and innovation in education.

2.2 National Strategic Development Plan

The Royal Government of Cambodia will roll out the National Strategic Development Plan (NSDP) 2024-2028 to contribute to the implementation of and closely align with the Pentagonal Strategy Phase 1 and to achieve the Cambodian sustainable development goal as well as Cambodia's Vision. However, this new phase document is still in a work-in-progress, and expect to release in 2024 or early 2025. The NSDP 2019-2023, the previous phase, was a key document guiding Cambodia's efforts in alignment with Rectangular Strategy Phase 4. It outlined strategies include (1) Acceleration of governance reform: the core of the Rectangular Strategy; (2) overarching environment for implementing the strategy; (3) human resource development; (4) economic diversification; (5) private sector and job development; and (6) inclusive and sustainable development. In terms of R&D, this document aimed to promote science and technology advancement, enhance productivity, promote innovation and enhance skills workforce underscoring the collaborative efforts with relevant national and international actors to advance R&D initiatives in the key sectors including agriculture, environment, green economy and so forth. One of the key highlight of the strategy is to enable effective and efficient research at the national level, covering the promotion of research, technology transfer, and the creation of research funds, and strengthening of research institutions, the implementation of research programmes, and the commercialisation of results.

2.3 Cambodia Sustainable Development Goals

The Cambodian Sustainable Development Goals (CSDGs), derived from the Global Sustainable Development Goal framework, address the efforts of the Royal Government of Cambodia to achieve sustainable development goals by 2030, coalescing the efforts with the Pentagonal Strategy and National Strategic Development Plan. Unlike the standard global SDGs, Cambodia has integrated its own unique 18th goal related to mine action and other explosive remnants of war in addition to the standard 17 goals. In terms of R&D, the strategy supports the integration of innovation and technology-driven solutions to achieve the goals such as Quality Education (Goal 4), Decent Work and Economic Growth (Goal 8), and Industry, Innovation, and Infrastructure (Goal 9), all of which are critical for advancing R&D efforts. It contains 105 targets, two of which have research embedded in the framework, one of which aims to increase investment in agricultural research to “end hunger, achieve food security, and promote sustainable agriculture” (CSDG 2), and another one is to enhance scientific research, increase the number of researchers, and promote public and private research to achieve CSDG 9: “Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation”. In addition to the overarching national strategies, several sectoral strategies are relevant for building national research capabilities.

3. The Sectoral Policies

There are a wide range of sectoral policies and regulation frameworks developed to promote and empower the development of each sector, which in general is in alignment with Cambodia’s national core policies. These policies directly and indirectly support the advancement of R&D efforts in the country. The objectives of these policies are to harness the potential of R&D to leverage innovative capabilities, foster economic development, and address the challenges faced by their sectors.

3.1 The National STI Policy 2020-2030

The National STI Policy, approved in 2019, aims to strengthen the national STI capabilities, including institutions and human resources, and improve the STI ecosystem. The strategies developed in the framework of the policy seek to:

- Develop STI human resources in terms of quantity, quality, and composition while accounting for ethics and gender equality;
- Develop an STI environment that maximizes the potential of human resources;
- Promote efficiency and effectiveness of R&D by adapting the use of technologies to the Cambodian needs and by learning from technologies being developed abroad;
- Develop a dynamic and robust innovation ecosystem able to synthesize technologies and engineering in priority industries; enable businesses to participate in international markets; and enhance productivity;
- Develop an STI culture to increase trust in the application of national technologies.

3.2 Cambodia's STI Roadmap 2030

Understanding the importance of Science, Technology and Innovation, the Government of Cambodia has established Ministry of Industry, Science, Technology & Innovation (MISTI) and the National Council of Science, Technology and Innovation (NCSTI) to oversee and give the direction of the efforts to advance STI in Cambodia. These two institutions are mandated, at the national level, to promote STI and lead STI transformation. They provide guidance to define a clear vision and objectives for STI strategies that enable the country to reach the goals set for 2030 and 2050. Currently, MISTI under guidance of NCSTI, has established and disseminated numerous policy and regulatory frameworks that drive STI transformation at the national level:

First and foremost, the Cambodia's STI Roadmap 2030, developed by MISTI and enacted in 2021, aims to support the implementation of the National STI Policy. This roadmap aims to strengthen national technological capabilities and improve innovation performance of Cambodia as these will be critical to achieve the ambitious vision of the Royal Government of Cambodia to become an upper-middle-income economy by 2030 and a high-income economy by 2050. Science, technology and innovation (STI) is considered to be a pivotal driver to shift the economic development pathway from a focus on traditional growth to support for inclusive and sustainable growth. ST| will enable and accelerate the structural transformations required to increase national prosperity, peace, security, safety and socioeconomic development and to improve quality of life. The roadmap is developed on five pillars, with one focusing specifically on research. The five pillars include:

- **Enhancing the governance of the STI system:** STI governance is key and has been recently restructured with the creation of MISTI in March 2020. It will be important to consolidate this new structure, while reducing fragmentation and breaking down silos. This will require clarifying the role of MISTI and other stakeholders, strengthen awareness and capacities of the Government to implement the National STI Policy, and monitoring and evaluating advances made in the promotion of STI.

- **Build human capital in STI:** Current demand for innovation is low and there is a limited scientific and entrepreneurship culture. It will be critical to promote scientific, digital and entrepreneurship literacy, and the technological readiness of the youth, starting in basic education. Teaching science, technology and innovation from very early age will help create a new generation of scientists and innovators. Skills in science technology engineering and mathematics (STEM) will also need to be promoted in higher education. In addition, there is room for strengthening teaching and collaboration with the private sector in technical and vocational education and training (TVET) institutions. Strategic development of human resources is at the foundation of promoting STI.
- **Strengthening research capacity and quality:** Build the capacity of the higher education and research system to conduct high quality R&D activities of national interest and in priority sectors is much needed. This will require developing a national research agenda with the academic community and in close collaboration with the private sector, providing funding to support excellent science and the internationalization of research and encourage collaboration with the private sector.
- **Increasing collaboration and networking between different actors**: Innovation comes from the exchange of ideas, across different people, organizations, sectors and scientific domains. Intermediary organizations and knowledge broker institutions are essential to facilitate such exchanges. Hence, it will be critical to promote and sustain incubation and acceleration facilities, technological platforms open to private sector and innovative clusters fostering collaboration to support innovation in small and medium-sized enterprises (SMEs) and enhance their absorptive capacities.
- **Fostering an enabling ecosystem for building absorptive capacities in firms and attracting investments in STI:** Supporting innovation capabilities and increasing the absorptive capacities of firms requires financing and promoting intermediary structures that nurture new firms (start-ups), support technology transfer and promote domestic technologies. It needs to be fostered by institutions that provide technology and quality (norms and certification) services to firms. It also requires

increasing access to finance for innovation activities, including through leveraging investments from the private sector and attracting funding from donors. Incentivizing foreign direct investment (FDI) that supports the building of domestic technological capabilities, facilitating the importing of technology equipment and promoting intellectual property rights are additional avenues for fostering an enabling ecosystem for innovation.

3.3 National Research Agenda

The **National Research Agenda 2025** of Cambodia, launched by the Ministry of Industry, Science, Technology, and Innovation (MISTI), represents a key step toward transforming Cambodia’s research landscape into a vibrant research ecosystem. The National Research Agenda is built upon the direction guided in the Cambodia’s STI Roadmap 2030. This agenda aims to guide research efforts toward high-priority sectors that will contribute to national goals such as sustainability, technological development, and economic resilience. By addressing essential areas—like energy, food security, digital health, education, and environmental sustainability—the agenda seeks to advance Cambodia's economic and social progress in line with its broader Vision 2030 and Vision 2050 goals.

The agenda identifies several core research priorities, each targeted to address specific development needs in Cambodia:

- **‘Local food’:** 70 percent of Cambodia food consumption is produced locally
- **‘Reliable Energy Supply’**: 90 percent of energy consumption is generated locally
- **‘Quality Education’**: Education meets international quality standards
- **‘Electronic and mechanical spare parts’**: Cambodia exports 70 percent of the electronic and mechanical spare parts produced in the country
- **‘Cloud-based services’:** Cambodia’s cloud-based services development is on par with ASEAN

- **'Electricity and potable water'**: All Cambodians have access to reliable electricity and safe potable water
- **'Carbon neutrality':** Cambodia becomes a carbon neutral country
- **'Digitally-enhanced health'**: All Cambodians have access to digitally-enhanced health services

The National Research Agenda 2025 serves as a comprehensive framework to mobilize resources, build institutional capacity, and foster partnerships across sectors. Its strategic objectives include:

1. **Investing in research to support the eight research missions**. This will include establishing a national research foundation and a national research fund to support investments in research and guide such investments towards the eight research missions.
2. **Strengthening the role and capacities of public research institutions** through the establishment of a national research system, national research fund, centers of excellence in research, and a national research publication platform.
3. **Supporting research careers.** This will require recognising the research profession, introducing a research career framework and establishing an attractive system of incentives for researchers to support research careers in the public and private sectors.
4. **Incentivising research activities and collaboration.** This will require enhancing coordination among research promoting institutions, setting up an adequate research management system; incentivising industry-academia-government and international collaborative research; building the absorptive capacity of firms and their ability to conduct R&D and innovation activities; establishing key infrastructure to support technology transfer and adoption; exploring means to incentivise R&D investments through tax incentives; and supporting greater enforcement of intellectual property rights.

These missions were co-created through a multi-stakeholder process and focus on research activities and resources on the achievement of national developmental goals, including those stated in the Rectangular Strategy Phase IV, the National Strategic Development Plan (NSDP) 2019-2023, and the Cambodia Sustainable Development Goals (CSDG) Framework for 2016- 2030.

3.4 Six Technology Roadmaps

MISTI has also produced and disseminated technology roadmaps on six priority sectors, guided and rectified by the formal meeting of NCSTI. These six priority sectors include Agriculture, Education, Health, Energy, Tourism, and Digital. These roadmaps are a comprehensive policy document navigating the efforts toward the dynamic landscape of technology in the priority sectors in the pursuit of socioeconomic development and a sustainable future. These roadmaps aim to utilize the power of Science and Technology to advance the six sectors to be more competitive and reliable, tackle national challenges, and most importantly significantly contribute to socioeconomic development. Moreover, these roadmaps serve as a cornerstone to help Cambodia achieve the 2050 vision. The roadmaps have specifically outlined the important role of R&D in achieving the goals. For example, in the energy technology roadmap, R&D is indicated to be very important in exploring new sources of energy and encouraged to invest more as it can strengthen scientific-based and technological capabilities and bring out more innovation.

3.5 Education Strategic Plan 2019–2023

MoEYS is implementing the Education Strategic Plan 2019-2023, which updates the 2014-2018 strategy, to support the three overarching national strategies, namely the NSDP, the RS-IV, and the CSDGs Framework. In terms of research, the overall objective of this Plan is to improve research quality; promote research among HEIs; foster continuous development through collaborations, internationalisation, and training; and provide financial support through fund mechanisms such as funds for research, scholarships, and training.

Specific objectives are defined for each education sub- section: Early childhood, primary, secondary, higher, non-formal, youth development, physical education, and sport. This policy aims to improve access to education for young Cambodians and enhance the quality of education services.

The Plan seeks to reinforce research among HEIs by supporting research on teaching methods and the development of research-based policies. The Government also aims to establish a research fund for higher education and to develop partnerships to improve research capacity. The Education Research Council is one of the mechanisms to reform academic management. It intends to guide research on education practices, and support policymakers and research institutions in designing policies and conducting research, respectively. The Plan also highlights the role of professors to conduct research on improving teaching methods, which, in turn can improve the quality of their work.

3.6 Policy on Higher Education Vision 2030

The Policy on Higher Education Vision 2030 establishes a long-term objective to improve the quality of higher education through good governance and mechanisms that ensure access to quality higher education. Broader access would fulfil the needs of the Cambodian labour market and respond to challenges related to socio-economic development. MoEYS also seeks to develop a programme to promote equity in access to higher education and to enhance skills development and knowledge excellence. Furthermore, the Policy aims to improve economic development through improved learning, teaching and quality of research systems, and to build a governance system that would allow for better management of relevant institutions. Regarding research, the Policy aims to increase the quality of research, promote collaborations with foreign universities, foster staff and students' engagement in the development of a national research culture, and strengthen research and innovation capacity.

3.7 Cambodian Information and Communication Technology Masterplan 2020

The Cambodian Information and Communication Technology (ICT) Masterplan 2020 aims to establish Cambodia among the top countries in South-East Asia on the ICT-related World Economic Forum indexes. The plan is divided into four main categories: empowering people, ensuring connectivity, enhancing capabilities, and enhancing e-services.

R&D governance is integrated as part of the enhancing capabilities section of the Masterplan. The Masterplan emphasises the need to develop the ICT industry and mentions that a research network is key for national ICT infrastructure to be developed further. To this end, it recommends creating pilot research networks in large cities. The Masterplan also notes the challenges by regarding the R&D environment, including the lack of research culture, R&D facilities, funds, equipment and, more generally, human resources.

3.8 Digital Economy and Society Policy Framework 2021-2035

The Government has recently developed a digital policy to address the challenges and opportunities emerging from the Fourth Industrial Revolution and crises such as the COVID-19 pandemic. The policy framework focuses on the digital transformation of the economy and society. It highlights the challenge of becoming more resilient towards the future which can be addressed through digital strategies while pointing out concerns regarding the development of databases and digital platforms. To transform itself into a digital economy and society, Cambodia will require an enabling infrastructure, reliability and confidence in the digital system, citizens, government, and businesses. The following are the actions of the policy framework focusing on research:

- Participate in cooperative and international digital research initiatives;
- Develop R&D capacity-building funds in ICT;
- Build the capacity of actors on digital adoption and transformation through research

and digital technologies;

- Foster digital research and innovation in HEIs, vocational training institutions and research centres;
- Foster R&D and innovation through partnership with businesses which would enable data and knowledge exchanges, including the development of a digital research and innovation centre and support funds;
- Strengthen the digitalization of documents in Khmer language to allow the identification of the language in search and facilitate data analysis;
- Develop a national strategy promoting R&D and innovation;
- Improve the quality of public research centres on digital R&D, develop a digital innovation fund to enhance businesses' investments in the sector, and strengthen policy implementation on STI to promote more conducive R&D and innovation environments.

3.9 Digital Economy and Society Policy Framework 2021-2035

The Cambodia Industrial Development Policy and Plan 2015-2025 is the main policy document focusing on the future of Cambodia's national industry. The Policy and Plan seeks to change the industrial structure by strengthening it, increasing the volume of exports, diversifying exported goods, and promoting SMEs. There are five priority sectors set by the Government:

1. New industries or manufacturing ventures with high value-added products, not only on consumer products but also machinery, mechanic, electronic, electric, and transport assembly, and natural resource processing;
2. SMEs in all sectors, especially in drugs and medical equipment, construction, packaging, furniture manufacturing, and industrial equipment;
3. Agro-industrial production for export and domestic markets;

4. Supporting industries for agriculture, tourism, textile, regional production chains linked to the provision of raw materials, especially for the garment sector, and production of spare parts and semi-finished products;
5. Industries supporting regional production lines of ICT, energy, heavy industries, cultural and traditional handicraft, and green technology.

On the aspect of research, the Industrial Development Policy and Plan 2015-2025 incentivises investment in research projects and proposes the creation of research funds to support this endeavour. The plan also seeks to strengthen research capacity according to the demand for industrial technologies. To improve industrial relations, it recommends exploring solutions to address industrial disputes. Regarding research at HEIs, it mentions the need to increase R&D quality by improving equipment quality and availability. Lastly, the plan proposes the creation of scientific and technological parks and a public research institute.

3.10 Agricultural Sector Strategic Development Plan 2019-2023

Agricultural Sector Strategic Development Plan 2019-2023 is developed and implemented by the Ministry of Agriculture, Forestry, and Fisheries. This policy is particularly aligned with the core policy Rectangular Strategy Phase 4 of the Government in the 6th legislature. The Plan focuses on different factors to improve agriculture in the future. The objectives relate to productivity, commercialization, and diversification of agriculture; animal health and production; aquaculture and fisheries development and management; sustainable forestry and wildlife resources; human resources and support services development and management. This policy has embedded R&D in its key elements within the development plans such as improving crop production, yield, quality, and resilience to climate change through developing better and resilient crop varieties, improving soil quality, and promoting sustainable farming practices. Furthermore, promoting horticulture such as in the areas of post-harvest technology and food safety, creating higher added value for the market in the region and globally.

3.11 Cambodia’s Circular Economy Strategy and Action Plan

This policy is initiated and rolled out by the Ministry of Environment in 2021, aiming to transition the country from a linear economy to a circular economy model to achieve its sustainable development goal and the fourth phase of the government’s rectangular strategy. The strategy puts a strong emphasis on sustainable resource use, resource efficiency, and waste reduction. This strategy outlines its key objectives to promote sustainable production and energy use, enhance waste collection and recycling, and recycling, and, importantly, foster environmental education. R&D plays an important role in this policy, such as creating effective and technology-based waste management solutions, particularly in waste recycling and renewable energy. Innovation, which can be enhanced by the R&D program, is encouraged in many areas such as eco-designed and sustainable materials. This strategy also aims to address the challenges, including a lack of infrastructure, regulatory gaps, and raise awareness and capacity building across sectors and relevant stakeholders.

3.12 Cambodia’s Climate Change Strategic Plan 2014- 2023

Cambodia’s Climate Change Strategic Plan 2014-2023 is developed by Ministry of Education as a country’s primary framework to address the climate change issues faced by many sectors and people in Cambodia through a low-carbon strategy and build climate resilient capacities across sectors, supporting sustainable development, and climate-oriented policies that align with national growth and global commitments to combat climate change. In this regard, this policy framework aims to address a gap in knowledge, capacity, and science-based decision-making in responding to climate change. This Plan aims to incorporate climate change into the existing national programmes and frameworks, and complement policy gaps by aligning itself with different sectoral climate change strategic plans developed by other line ministries and national agencies to reduce vulnerability and increase public participation in shifting to low-carbon development. This strategic plan has embedded R&D in its strategic plan, particularly in fostering resiliency and adaptation through innovative technology. The Plan is structured into three phases: immediate, medium, and long-term, with research being the focus of the third phase (2019-2023).

3.13 Third Health Strategic Plan 2016-2020

The Ministry of Health (MoH) developed the Third Health Strategic Plan 2016-2020 as a strategic management tool to guide health institutions in managing their resources and implementing health strategies. Research is one of the strategic objectives described in the Plan: it seeks to promote research in the health sector, increase research capacity, and improve available ICT infrastructure for research. Health information system is a key strategic objective aiming at enhancing the reliability and accuracy of data and information that can promote health research. This strategic objective outline the activities to promote a better use of research findings.

3.14 New Law on Investment

The Council for the Development of Cambodia drafted the new investment law in February 2021, and it was promulgated in 15 October 2021.The Law aims to develop a legal framework to enhance domestic and foreign investment based on principles of transparency, predictability, and favourability. This new law, through the generous incentive schemes, aims to attract and promote quality, effective and efficient investments in specific sectors that have a high potential contributing to socio-economic development, including high-tech industries conducting R&D and innovation activities. Under this Law, R&D and innovation activities are entitled to investment incentives. The incentives are either in the form of income tax exemption (for up to 9 years) or special depreciation (up to 200% of specific expenses) for the Qualified Investment Project (QIP), There are also additional incentives and special incentives for any specific sector and investment activities that deems having a high potential that contributes to Cambodia’s national development.

The following investment sectors and activities are entitled to receive the investment incentives, including:

1. High-tech industries involving innovation or R&D
2. Innovative or highly competitive new industries or manufacturing with high added value
3. Industries supplying regional and global production chains
4. Industries supporting agriculture, tourism, manufacturing regional and global production chains and supply chains
5. Electrical and electronic industries
6. Parts, assembly and installation industries
7. Mechanical and machinery industries
8. Agriculture, agro-industry, agro-processing and food processing industries serving domestic market or export
9. SMEs in priority sectors and cluster, industrial parks, and STI parks
10. Tourism
11. Special economic zones
12. Digital industries
13. Education, vocational training and productivity promotion
14. Health
15. Physical infrastructure
16. Logistics
17. Environmental management and protection, biodiversity conservation and circular economy
18. Green energy, technology contributing to climate change adaptation and mitigation
19. Other sectors deemed by the Royal Government of Cambodia to have potential for socio-economic development

4. Conclusion

In conclusion, Royal Government of Cambodia has prioritized scientific and technological progress as the catalyse to promote socio-economic development and to tackle the national challenges, and to achieve the Cambodia's Sustainable Development Goals, in align with the global (SDG), and most importantly to realize the government's ambitious vision to transform current Cambodia's economy, labor's based, to become upper-middle income country by 2030 and high-income country by 2050, skill-based and innovation led economy. To achieve these goals, Research and Development is the cornerstone for the socio-economic development, enhancing innovation capacities, promoting start-up and entrepreneurship, strengthening competitiveness, developing skill-based labor, and enhancing innovative solutions to tackle national challenges such as climate change and global warming. These policies provide a foundational framework that the Cambodia's R&D Management System, initiated by Ministry of Industry, Science, Technology and innovation, can build upon, especially in advancing digital and technological innovation, resource allocation, and cross-sectoral collaboration. Policies such as the National Science, Technology, and Innovation Policy, Cambodia's STI Roadmap 2030, National Research Agenda, particularly the sectoral polices overarching prioritized sectors of Cambodia including the Agricultural Sector Strategic Development Plan, and the Circular Economy Strategy reflect Cambodia's commitment to sustainable development and economic resilience, emphasizing a need for cohesive R&D efforts across sectors.

For the Cambodia's R&D Management System, these policies indicate a strategic advantage in terms of leveraging governmental priorities and resources to support data collection and analysis, knowledge creation and sharing, and promote both horizontal and vertical technology transfer within the National Innovation System. With a systematized and digitalized framework, the system could effectively facilitate and support the ongoing efforts in agriculture, environmental sustainability, and climate adaptation by promote centralizing data and enabling efficient resource allocation and policy monitoring.

Additionally, by aligning with Cambodia's sustainable development priorities, the system can provide a platform for public-private partnerships, international collaboration, and investment in priority areas, addressing resource limitations and enhancing the efficiency resource utilization with transparency.

Successful implement and integrate this R&D Management System can further strengthen the research capacity and quality of Cambodia's researchers, particularly the address the national challenges and leverage STI to achieve country's socio-economic development as indicated in Cambodia's STI Roadmap 2030, and National Research Agenda 2025. Furthermore, this digital platform can enhance Cambodia's capacity for innovation, support efficient policy implementation, and contribute to achieving national goals outlined in strategic plans such as the Pentagonal Strategy Phase I, National Strategic Development Plan and the Cambodia Sustainable Development Goals.

CHAPTER 4

Cambodia's National R&D Information Management System (CTIS) Concept and Development Plan of Actions Service

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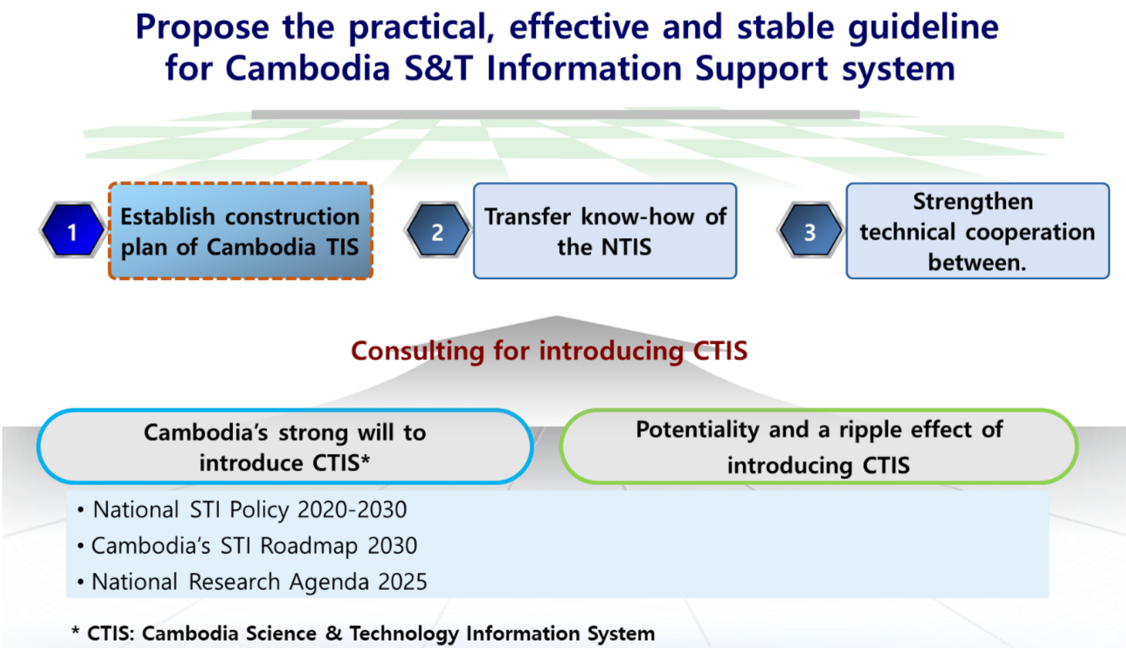
Chapter 4. Cambodia's National R&D Information Management System (CTIS) Concept and Development Plan of Actions Service

1. Introduction

1.1 Background and Objectives

Cambodia recognizes the critical role of R&D in driving innovation and promoting her competitiveness in the region and globe for the economic and sustainable development, as clearly indicated in the national policies including Cambodia's Science, Technology and Innovation Roadmap 2030 and National Research Agenda 2025. To fully utilize the potential of R&D, it is required a comprehensive information and management system to be in place. In addition, Cambodia has also identified the need to transition to e-government as a long-term ambitious vision, illustrated in its S&T innovation policy and the implementation of its top-level national development strategy, the Pentagonal Strategy, and has shown its steadfast in the commitment in capacity building through efficient and effective national R&D and technology transfer initiatives. Therefore, MISTI and STEPI, with the technical support from KISTI, which has the know-how and experiences in building a national R&D information management system and NTIS, will establish a target system and action plan to build a Cambodia's National R&D Information Management System/Platform (CTIS) that is in line with all relevant Cambodia's policies and utilizes the accumulated information.

[Figure 4-1] Background and Purpose



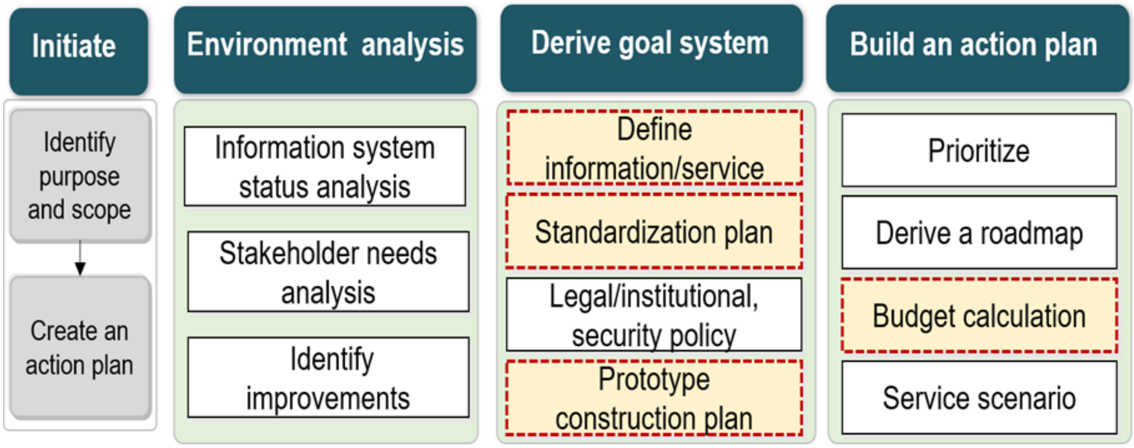
Source: Author's elaboration based on KISTI NTIS Center, 「Vietnam FS study」.

1.2 Business Scope

This report covers the typical pre-feasibility study process of defining information and services, planning to build a prototype, and calculating a budget.

- **Define Information/Service:** Define main services and information according to the current state of Cambodia's National R&D Information Management System.
- **Standardization Plan:** Establishment of National R&D information standard items for joint use and management of R&D information.
- **Prototype Construction Plan:** Derive a target system suitable for Cambodia and establish a plan for implementation.
- **Budget Calculation:** Estimate the cost to build the planned target system.

[Figure 4-2] Business Scope



Source: Author's elaboration.

1.3 Promotion Strategy

As an implementation strategy for the development of a national R&D information management prototype in Cambodia, we analyze the results of interviews with relevant organizations in Cambodia in 2023, establish a construction plan based on MISTI needs, derive target system deliverables, and calculate construction costs. Our approach the entire process is being carried out in cooperation with MISTI Cambodia.

[Table 4-1] Strategy by action plan

Plan	Method
Cambodia R&D Information Management System Research/Analysis	• Research and analysis of R&D information management status in Cambodia through relevant organizations (Survey on National R&D Management System for Cambodia, 2023) • National needs
Derivation of Target system	• Derivation of key tasks and goal system for Cambodia's R&D information management system, reflecting Cambodian national R&D information management system status analysis results, national needs, and MISTI opinions
Derivation of Construction Plan	• Derivation of roadmap and construction cost based on Cambodia's R&D information management target system

Source: Author's elaboration.

2. Cambodia R&D Information Management System Research/Analysis

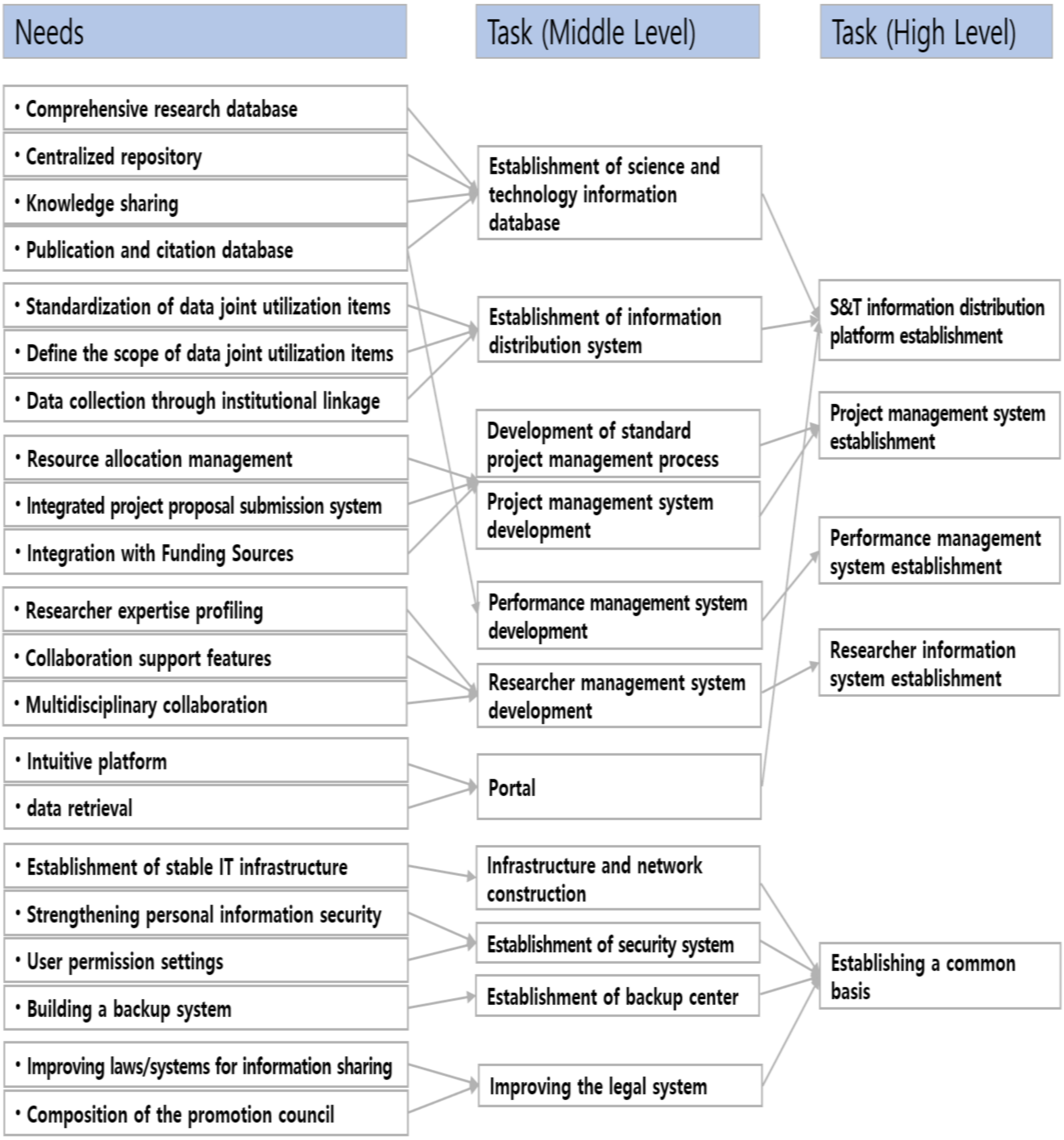
2.1 R&D Information Management System Needs Analysis

Based on the results of the Cambodian National R&D Information Management Status Survey interviews and scientist community demands, we collected and analysed the pressing needs of the system tailoring to Cambodia's context, and categorized them into information systems, laws/institutions, and standardization, and derived five key functionalities based on their characteristics and similarities.

The main demands were for an integrated research database, standardization of jointly used information, project management (proposal, execution, evaluation, etc.), researcher information, data search/portal, stable infrastructure, security, backup system, and legal/institutional improvements. Based on this, we derived mid-level tasks such as Establishment of science and technology information database, Establishment of information distribution system, Development of standard project management process, Project management system development, Performance management system development, Researcher management system development, Portal Establishment, Infrastructure and network construction, Establishment of security system and backup center, Improving the legal system.

We grouped the middle tasks into similar ones and finally came up with five action tasks: Science technology information distribution platform establishment, Project management system establishment, Performance management system establishment, Researcher information system establishment, and Establishing a common basis.

[Figure 4-3] R&D Information Management System Needs Analysis in Cambodia



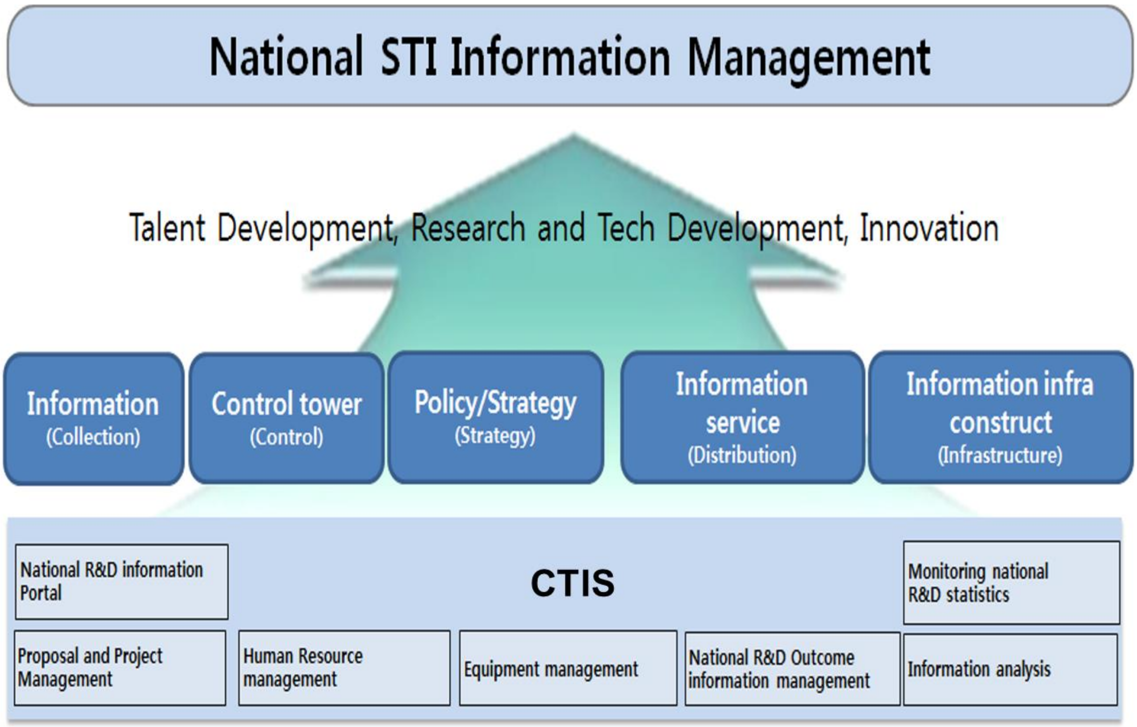
Source: Author's elaboration.

3. Promotion Plan

3.1 Vision and Target system

The vision is to build a national R&D project management system (proposal, execution, evaluation, etc.), researcher information management system, equipment management system, etc. as prototypes, and to manage national S&T information through information collection, policy formulation, and information service provision based on Cambodia's integrated national R&D information system.

[Figure 4-4] Cambodia National R&D Information System Vision

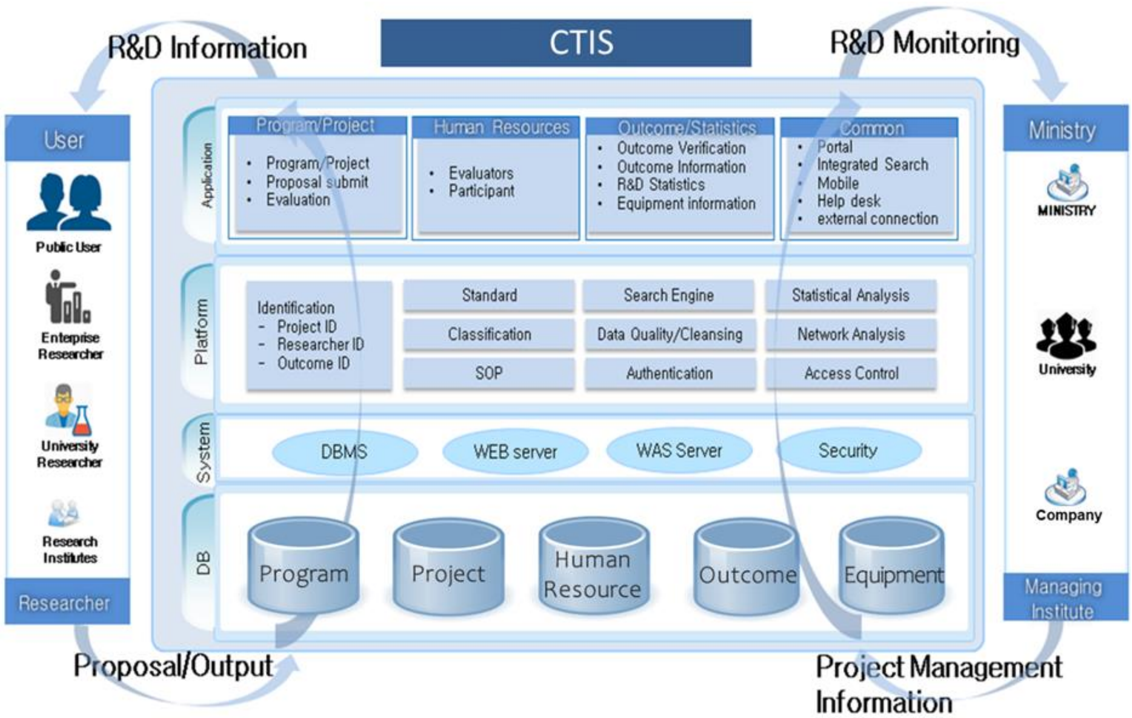


Source: Author's elaboration based on Development on Platform Model for National R&D Information of Costa Rica (2016), KISTI Report.

3.2 Services

The figure below is a conceptual view of the Cambodian National R&D Information Integration System. The Cambodia National R&D Information Common Platform will be able to collect, link, and service information on projects, tasks, researchers, outcomes, etc. from organizations conducting national R&D projects. The system will provide an integrated database, information system, platform, and information services from the perspective of ministries, agencies, and researchers in Cambodia's national R&D management. After building a prototype, you'll need to expand your collection/linkage organizations, data items, and services offered.

[Figure 4-5] Target System Concepts



Source: Author's elaboration based on Development on Platform Model for National R&D Information of Costa Rica (2016), KISTI Report.

- CTIS Portal: A unified portal to provide integrated search services for national R&D information, including programs, projects, researchers, achievements, and research facilities/equipment.
- National R&D project management: Provides current information on national R&D programs and projects, including project summary information (classification, keywords, etc.), participating researchers, induced outcomes, and evaluation.
- Researcher Information management: Provides information on the history (Education/Career, Project, Paper, Patent, Books, Major Activities, Awards) of researchers who participated in National R&D projects
- R&D Outcome Information management: Provides outcome information such as papers and patents generated by National R&D project.
- Facilities and equipment information management: Provides Research Facility & Equipment Search function, Research Facility & Equipment Information, Research Facility & Equipment Reservation service.
- National R&D announcement: provides national R&D announcements from more than 120 ministries and project management institutes. And user can Register the ministries or keywords of users' interest and conveniently receive public announcements by email.
- Statistical information service: provides domestic and foreign science and technology statistical information such as major science and technology statistics, technology capability evaluation, science and technology prediction, and various science and technology indicators related to research facilities and equipment.

3.3 Information

Basic items proposed for national R&D information collection/linkage for Cambodia's National R&D Information Management Prototype System. These can be changed, deleted, or added based on the Cambodia's environment. For example, as shown in item 9 of the project basic information, 'Continuous Project Y/N' means whether the project is classified/secure or not. If the concept of a security assignment does not exist, you can confirm by deleting the item.

[Table 4-2] Information items

Category	Information Item	Number	Name
Project Information	Program	1	Program Name
		2	Name of Ministry
		3	Program Period
		4	Program Budget
		5	Funding Budget
		6	Enforcement Amount
		7	Budget Amount of Agreement
		8	Program purpose classification code
	Basic Information	9	Continuous Project Y/N
		10	Project ID
		11	Project Title
		12	Total project period
		13	Research Subject Code
		14	Region
		15	Standard Classification Code
		16	6T Technology related Code
		17	Summary of Research Objectives
		18	Summary of Research
		19	Expected Effect Summary
		20	Khmer Keyword
		21	English Keyword
		22	Confidential Project Y/N

Category	Information Item	Number	Name
		23	Confidential Project End Date
		24	Period of the Project
		25	Project Management Organization Name
		26	Name of Organization
		27	R&D Phase Code
		28	Project Progress Status
		29	Government Investment Research Budget
		30	Data of Signing
	Final Evaluation Result	31	Project Identification Number
		32	Aborting the Project Y/N
		33	Type of the Abortion
		34	Reason for the Abortion
		35	Abortion Data
		36	Final Evaluation Result_Evaluation Data
		37	Final Evaluation Result_Rating
		38	Final Evaluation Result_Score
		39	Final Evaluation Result_Evaluation Category
	Evaluator	40	Project Identification Number
		41	Date of Participation Number
		42	Science and Technology Registration Number
		43	Name of the Evaluator
		44	Evaluation Step
	Project Budget	45	Personnel Expenses(Salary)
		46	Direct Expenses
		47	Overhead
		48	Budget for Commissioned Research Project
		49	Matching Fund
	Commissioned Projects / Joint Research	50	Co-operative Research Y/N
		51	Commissioned/Co-operative Research Project Number
		52	Commission Project Name
		53	Name of Performing Organization
		54	Commissioned Project_Total Research Period
		55	Commissioned Project_Total Research Budget

Category	Information Item	Number	Name
	Participation Researcher	56	Role Code
		57	Researcher Number
		58	Participation Period
		59	Participation Rate
		60	Participation Researcher Gender
		61	Participation Researcher Major
		62	Participation Researcher Degree
		63	Number of Participating Researchers
	Project Announcement	64	Name of Ministry
		65	Name of the announcing Agency
		66	Notice Title
		67	Date of Announcement
		68	Application Deadline
		69	Announcement Contents
		70	Application URL
Outcome Information	Publication	71	Journal Title
		72	Publication Title
		73	ISSN_ISBN
		74	Major Author Name
		75	Journal Volume Number
		76	SCI Code
		77	Co-author Names
		78	Domestic or Foreign
		79	Abstract
		80	Conference Hosting Country Code
		81	Conference Paper Title
		82	Date of Publication
		83	Start Page Number
		84	End Page Number
		85	DOI
	Intellectual Property	86	Application or Resister
		87	Title of the Patent
		88	Application Country Code
		89	Application Number

Category	Information Item	Number	Name
		90	Registration Number
		91	Date of Application
		92	Applicant Researcher Number
	Research Report (Result)	93	Report Title
		94	Issuing Organization Name
		95	Issuing Country Code
		96	Issue Date
		97	Used Language
		98	Whether the source is disclosed
		99	Abstract
		100	Keywords
		101	Management Organization Name
Researcher Information	Career	102	Place of Work(Career)
		103	Title
		104	Employment Period
	Basic Information	105	Name of Participant
		106	Institution Classification
		107	Mobile Phone Number
		108	Specialty
		109	E-Mail
		110	Researcher Number
	Publication	111	Publication Date
		112	Khmer Title
		113	English Title
		114	Journal Name
		115	ISSN_ISBN
		116	Major Author Name
		117	Journal Volume Number
		118	SCI Code
		119	Co-author Names
		120	Domestic or Foreign
		121	Abstract
		122	Conference Hosting Country Code
		123	Conference Paper Title

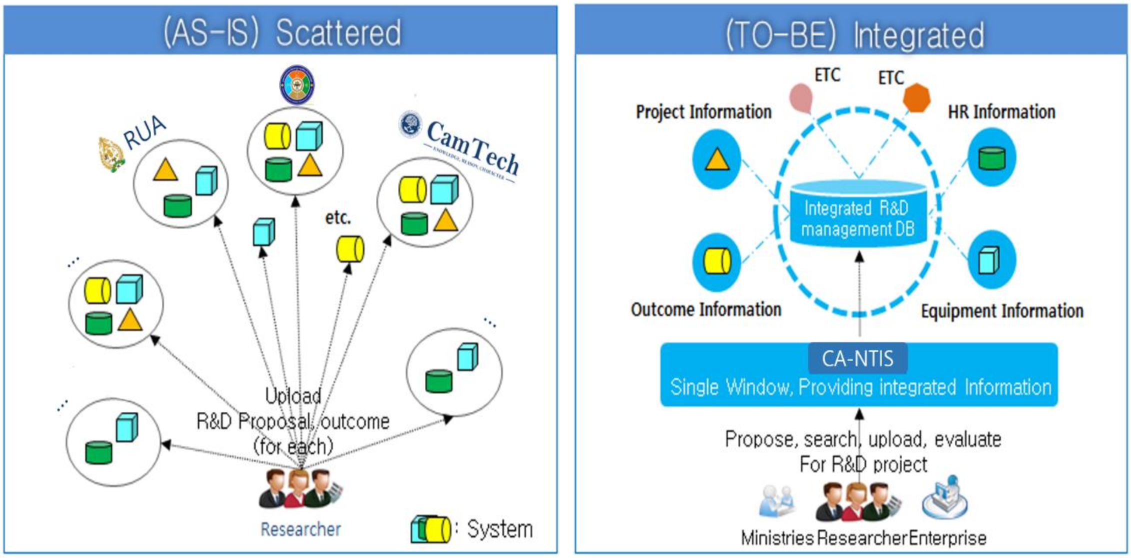
Category	Information Item	Number	Name
		124	Date of Publication
		125	Start Page Number
		126	End Page Number
		127	DOI
	Work Experience	128	Address of the Organization
		129	Office Phone Number
		130	Department
		131	Institution Name
		132	Title
		133	First Appointment Date
	Education Qualification	134	Dissertation Title
		135	Degree Date
		136	Major
		137	Acquisition University
		138	Degree
		139	Admission Date
		140	Year of Graduation
		141	Department
		142	Country Code
		143	Name of the Advisor
Evaluation committee	Basic Information	144	Expert Classification of each Ministry
	Committee	145	Responsibility
		146	Position
		147	Name of the Committee
	Book Writing (Translation)	148	Translator name
		149	Publication year
		150	Publisher
		151	Writing or Translation
	Award	152	Date of Award
		153	Name of Granting Organization
		154	Name of Award

Source: Author's elaboration.

3.4 Information Linking Institutes

In the environment where Cambodia is conducting national R&D projects or operating information systems for each individual institution, it should be built in the direction of overall integration for work efficiency and integrated DB utilization.

[Figure 4-6] Integrated national science and technology database



Source: Development on Platform Model for National R&D Information of Costa Rica (2016), KISTI Report.

It is necessary to identify the number of major institutions related to the Cambodian national R&D project and the number of key information such as tasks and performance of the institutions determined to be targeted for the prototype linkage, as well as the DB structure for integration.

[Table 4-3] Data holdings by organization

Information	Project	Researcher	Paper	Patent	Total
Institute of Technology of Cambodia	-	-	-	-	-
Royal University of Phnom Penh	-	-	-	-	-
Royal University of Agriculture	-	-	-	-	-
University of Health Science	-	-	-	-	-
Ministry of Education	-	-	-	-	-
Ministry of Civil Service					
Ministry of Labor					
Ministry of Foreign Affairs and International Cooperation					
Ministry of Commerce					
Ministry of Industry and STI					

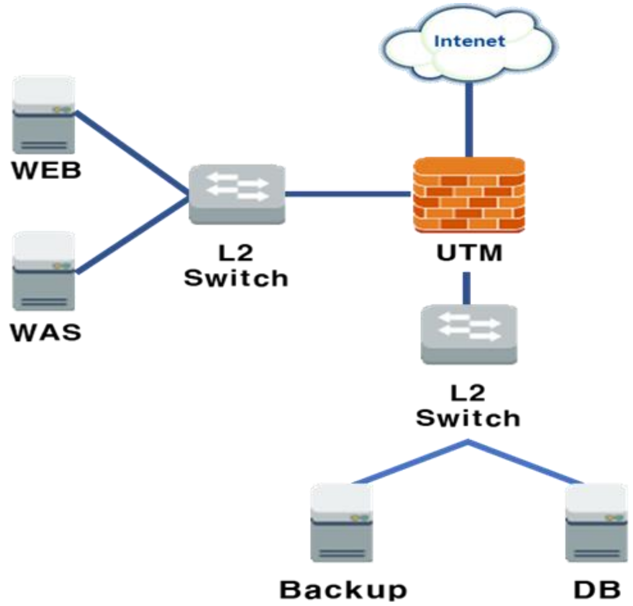
Source: Author's elaboration.

3.5 Infrastructure

Currently, Cambodia has a well-established infrastructure environment for building and operating a prototype-level information system, so we would like to organize the following specifications within the already established infrastructure environment without introducing new equipment.

[Figure 4-7] Infrastructure concept diagram

Infrastructure diagram



[Table 4-4] Infrastructure specifications and availability

Name	Specification	Availability
UTM	10/100/1000 Base-T 8Port	All next-generation firewall features are already existing in the infrastructure 1Gbps speed is not available, all connections are using fiber optic, minimum speed is 10Gbps
	1000Base-X 4Port	
	Firewall [Essential]	
	IPS [Selection]	
	SSLVPN [Selection]	
DMZ L2 Switch	10/100/1000 Base-T 24Port	L2 Switches are not available. L3 Core Switches, Access Switches are used in the infrastructure with uplink speed of 100Gbps, 40Gbps, 25Gbps and 10Gbps
	1000Base-X 4Port	
INT L2 Switch	10/100/1000 Base-T 24Port	
	1000Base-X 4Port	
WEB/WAS Server	CPU : Intel Xeon-Gold 3.6GHz 16core	Allocable
	Memory : 64GB	
	Disk : 1.92TB SSD * 2EA	
	OS: Free [CentOS, Rocky Linux etc.]	
	Apache, Tomcat	

Name	Specification	Availability
DB Server	CPU : Intel Xeon-Gold 3.6GHz 16core	Allocable
	Memory : 64GB	
	Disk : 1.92TB SSD * 2EA	
	OS : Free[CentOS, Rocky Linux etc.]	
	DB : Free[mongo, Maria, Postgre etc.]	
Backup	CPU :Intel Xeon-Silver 2.8GHz 16core	Allocable
	Memory : 64GB	
	Disk : 960TB SSD * 2EA 4TB NLSAS * 4EA	
	OS: Free [CentOS, Rocky Linux etc.]	
	Backup S/W inclusion	

Source: Author's elaboration.

3.6 Budget

To calculate the budget for the prototype of the Cambodian National R&D Integrated Information System, we used the same methodology as the budget for the system in Korea. The budget was calculated based on the software cost calculation guide published by the Korea Software Industry Association in 2024. In detail, we calculated the required investment for each major service by reflecting the labor and expenses of developers, consultants, and designers published in the software cost calculation guide. As the actual system will be built by a Korean system development company that received KISTI's technology, the budget is calculated using the wages of Korean SW worker. The M/M used in budgeting represents the amount/scope of work that one person should do in a month. In other words, it is meant to express how many technicians are needed to develop a piece of software. It is a concept used to understand the scope and cost of development.

[Table 4-5] Estimated costs for major groups

Group	Budget	Priority
Consulting (2 months)	$22,000	1
Information system construction/operating (H/W, network, etc.)	-	1
Design, Publishing	$14,500	1
National R&D information portal development (prototype)	$7,850	1
Project management	$47,100	1
Initial DB construction National R&D access control	($) 15,700	
Program/Project/Proposal/Evaluation	($) 31,400	
Outcome management	$47,100	1
Initial DB construction National R&D access control	($) 15,700	
Paper, Patent, Final report	($) 31,400	
Researcher management	$47,100	1
Initial DB construction National R&D access control	($) 15,700	
Education/Career, Participation projects, etc.	($) 31,400	
Tax, Staying expenses (Withholding tax, Business trip expenses, etc.)	$65,000	1
Subtotal (Priority 1)	$250,650	
Announcement management (Registration, inquiry, change, deletion)	$15,700	2
Facility/Equipment Management	$47,100	2
Registration, inquiry, change, deletion, location	($31,400)	
Reservation	($15,700)	
Statistics information	$47,100	2
R&D expenditures, number of projects (by year, by classification, etc.)	($11,775)	
Research and development researcher (by region, by classification, by major, etc.)	($11,775)	
Outcome (Papers, Patents, Facilities/Equipment)	($11,775)	
OECD Statistics	($11,775)	
Total (Priority 1,2)	**$360,550**	

Source: Author's elaboration.

- Expected Cost (Priority 1): $ 250,650
  - Consultant: 2M/M * $11,000 (Including financial and technical costs) = $22,000
  - System Software Developer: 19 M/M * $7,850 (Including financial and technical costs) = $149,150
  - Designer: 3M/M * $7,250 = $14,500
  - Withholding tax, Business trip expenses = $65,000
- Expected Cost (Priority 1, 2): $360,550
  - Consultant: 2M/M * $11,000 (Including financial and technical costs) = $22,000
  - System Software Developer: 33 M/M * $7,850 (Including financial and technical costs) = $259,050
  - Designer: 3M/M * $7,250 = $14,500
  - Withholding tax, Business trip expenses = $65,000

4. Expected Outcomes

4.1 Technical aspects

- Establishment and operation of system infrastructure through systematic connection and integration of national R&D information and standardization of distribution
  - Securing a science and technology information distribution base that can efficiently collect, manage, and utilize heterogeneous information distributed across research management specialized organizations by institution
- Comprehensive service of national research-based information and science and technology information for systematic management and utilization of national R&D information

4.2 Economic Outcomes

- Improving national R&D efficiency and productivity
  - Open research management and support through comprehensive analysis of national R&D status and sharing and dissemination of results
  - Enhancing researchers' research productivity and expanding economic and industrial ripple effects through systematic national R&D management and support
- Improving the management and operation efficiency of national R&D information resources
  - Reduction of information infrastructure fixed costs by securing interoperability and compatibility of all R&D resources subject to joint utilization, connection, and integration, such as H/W, S/W, and data

4.3 Social and cultural aspects

- Establishing a culture of establishing strategies and developing policies at the national level through mutual cooperation across all ministries
- Forming social and cultural consensus through mutual cooperation between industry and academia to strengthen national R&D capabilities from the perspective of the national innovation system.

4.4 User aspect

- User
  - Contribute to the popularization of science and technology by providing easy access to scientific and technological knowledge and information and improving understanding of the status of national R&D projects.
- Industry – Academia - Research officials
  - Support for research activities and utilize information through the use of reliable national R&D research results
  - Solve the problem of insufficient manpower and equipment by jointly utilizing the manpower and equipment owned by universities and research institutes.
- Government
  - Analysis of various information for national R&D policy development and securing transparency of business progress.

CHAPTER 5

Science &Technology Classification and its implication for Cambodia

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Chapter 5. Science & Technology Classification and its implication for Cambodia

1. Introduction

1.1 Concept and purpose

Science and technology standard classification system refers to a standardized framework that can systematically classify and organize the nation's science and technology activities, which may include human resources of researchers, engineers, professors, and students, R&D programs and projects, R&D equipment, and R&D outputs of journal papers and patents. For the effective operation of a national R&D information management system, these information and activities should be classified and organized systematically so that it can be used as a basic tool for policy makers to manage and evaluate research and development (R&D) activities, technological innovation, and related policies effectively.

The main purpose of this classification system for a country can be

a. A guideline for establishing a country's science and technology policies including R&D monitoring, R&D investment decisions, and performance evaluation,

b. A guideline for efficiently distributing and managing national R&D resources,

c. To compare the scientific and technological levels of a country and the comparable countries in international scientific and technological cooperation and to explore opportunities for international cooperation based on this.

1.2 Why each country needs its own system

Since the level of scientific and technological development, industrial structure, and key research areas of each country are different, a unique classification system for each country that can accurately reflect these issues is desirable and necessary. What is the main purpose of Cambodia MISTI's national R&D information management system? Is it more interested in efficient allocation of R&D resources into basic science and technology fields? Or is it more interested in industry-academia R&D collaborations? Industry tends to focus on new product developments and process improvement and academia tends to focus on scientific and technical field that support Industry's innovation activities.

Therefore, in terms of the national R&D management framework, it is paramount to establish a consensus on Cambodia's own science and technology classification system. Cambodia's MISTI can learn from how international organizations and advanced countries in the field of science and technology are utilizing their own science and technology classification system for their own policy purposes. In this regard, we explore three different classification systems with different purposes: one case from the Republic of Korea for a national government perspective, "National Science and Technology Standard Classification System", one case from Japan of a National R&D program perspective, "KAKENHI Review Section Table", and another case from OECD from an international organization perspective, "Field of Science and Technology" and "OECD Taxonomy of Economic Activities Based on R&D Intensity", respectively.

2. Science and Technology Classification Systems Environment and Practices

2.1 National Science and Technology Standard Classification System (Republic of Korea)

The National Science and Technology Standard Classification System in Korea is a science and technology standard classification system that the Korean government created for efficient management of science and technology-related information, human resources, research and development projects, research planning, evaluation, and management of national research and development projects, science and technology forecasting and technology level evaluation, and management and distribution of science and technology information at the national level.

The system is governed by the Science and Technology Basic Act. The Article 27 of the Act (Establishment of the National Science and Technology Standard Classification System) and Article 41 of the Enforcement Decree of the Act specify the 'standard classification.' The law also stipulates that the standard classification system should be actively utilized in,

1) Research planning, evaluation, and management of national R&D programs and projects,
2) Science and technology forecasting and technology level assessment, and
3) Management and distribution of science and technology knowledge and information with the purpose of efficiently managing information, human resources, and R&D programs and projects related to science and technology.

The Minister of Science and ICT (MSTI), Republic of Korea is the ministry in charge and the Korea Institute of Science and Technology Evaluation and Planning (KISTEP) has been designated as the agency in designing and improving the National Science and Technology Standards Classification System. Korea's science and technology standard classification was first enacted in 2002, and it was revised every three years since its enactment in 2002. Then, the revision cycle had been changed to every five years from 2012, and it is specified that revision cycles for detailed areas will be implemented twice a year (third and fifth years) starting from 2023.

The most recently revised version of the National Science and Technology Standard Classification in Korea is as of April 6, 2023. After the 2023 revision, the current classification system is composed of 22 major research fields and 2 application fields (public sector and industry sector). In case of 22 research fields, they can be divided into 277 two-digit categories with 2,799 sub-categories. In case of 2 application fields, there are 33 three-digit categories 9 (13 categories for public sector and 20 categories for industry sector).

[Table 5-1] Major Categories in Research Fields of the National Science and Technology Standard Classification System in Korea (revision in 2023)

Field	Major Category	Field	Major Category
N. Nature	NA. Math	H. Humanities and sociology	HF. Humanities
	NB. Physics		
	NC. Chemistry		HG. Social Sciences
	ND. Earth Science		
L. Life Science	LA. Biology		HH. Cultural Arts and Physical Education
	LB. Food for agriculture, forestry, and Fisheries		
	LC. Health care	O. Human Science and Technology	OA. Brain science
E. Artifact	EA. Mechanics		
	EB. Materials		
	EC. Chemical Engineering		OB. Cognitive/Emotional science
	ED. Electric/Electronics		
	EE. Information and Communication		
	EF. Energy/Resources		OC. SnT and Humanities and social sciences
	EG. Atomic		
	EH. Environment		
	EI. Construction/Transport		

Source: Lee, S. et al. (2023)

2.2 National Research Program: Japanese KAKENHI case

Grants-in-Aid for Scientific Research (KAKENHI, 科学研究費助成事業) are one of research fund programs that supports all scientific research proposed by researchers in Japan. As presented in [Figure 5-1], KAKENHI is intended to provide funds to curiosity driven researches with creative ideas from scientific researchers based on competitive selection process. They cover from basic to applied research in all fields, ranging from the humanities and the social sciences to the natural sciences as shown in [Table 5-2]. Since Fiscal Year 1999, some of the Grants-in-Aid program were delegated to Japan Society for the Promotion of Science (JSPS, 日本学術振興会) from the Ministry of Education, Culture, Sports, Science and Technology (MEXT, 文部科学省).

[Figure 5-1] Positioning of KAKENHI program in the policy on the promotion of science, technology and scientific research in Japan

Research type / Funding type	Scientific research based on researcher's creative ideas 【curiosity-driven research】	R&D on policy imperatives 【mission-oriented research】
Competitive Research Funds (Selected through open calls and review)	**Research supported by Grants-in-Aid for Scientific Research**	Research funded by open call and selection in line with the missions set by individual Ministries
Basic funds (Provision of management and operation funds, etc.)	Research conducted at universities and inter-university research institutes	National projects led by the initiative of Government, strategically promoted R&D projects conducted by National Research and Development Agencies

Source: JSPS, KAKENHI Pamphlet 2024, https://www.jsps.go.jp/file/storage/kaken_pamph_e2024/kakenhi_e2024.pdf, accessed 17/10/2024

As shown in [Table 5-2], the Review Section Table by Grants-in-Aid for Scientific Research-KAKENHI has eleven main categories of broad section with 64 sub-categories of medium-sized section.

[Table 5-2] The Review Section Table by Grants-in-Aid for Scientific Research-KAKENHI

Broad Section	Medium-sized Section
A. Humanities and Sociology	1. Philosophy, art, and related fields
	2. Literature, linguistics, and related fields
	3. History, archaeology, museology, and related fields
	4. Geography, cultural anthropology, folklore, and related fields
	5. Law and related fields
	6. Political science and related fields
	7. Economics, business administration, and related fields
	8. Sociology and related fields
	9. Education and related fields
	10. Psychology and related fields
B. Math, Physics, and Earth Science	11. Algebra, geometry, and related fields
	12. Analysis, applied mathematics, and related fields
	13. Condensed matter physics and related fields
	14. Plasma science and related fields
	15. Particle-, nuclear-, astro-physics, and related fields
	16. Astronomy and related fields
	17. Earth and planetary science and related fields
C. Engineering	18. Mechanics of materials, production engineering, design engineering, and related fields
	19. Fluid engineering, thermal engineering, and related fields
	20. Mechanical dynamics, robotics, and related fields
	21. Electrical and electronic engineering, and related fields
	22. Civil engineering and related fields
	23. Architecture, building engineering, and related fields
	24. Aerospace engineering, marine and maritime engineering, and related fields
	25. Social systems engineering, safety engineering, disaster prevention engineering, and related fields

Broad Section	Medium-sized Section
D. Chemical and Material Science	26. Materials engineering and related fields
	27. Chemical engineering and related fields
	28. Nano/micro science and related fields
	29. Applied condensed matter physics and related fields
	30. Applied physics and engineering and related fields
	31. Nuclear engineering, earth resources engineering, energy engineering, and related fields
	90. Biomedical engineering and related fields
E. Basic Chemistry	32. Physical chemistry, functional solid state chemistry, and related fields
	33. Organic chemistry and related fields
	34. Inorganic/coordination chemistry, analytical chemistry, and related fields
	35. Polymers, organic materials, and related fields
	36. Inorganic materials chemistry, energy-related chemistry, and related fields
	37. Biomolecular chemistry and related fields
F. Agriculture and Aquatic Science	38. Agricultural chemistry and related fields
	39. Agricultural and environmental biology, and related fields
	40. Forestry and forest products science, applied aquatic science, and related fields
	41. Agricultural economics and rural sociology, agricultural engineering, and related fields
	42. Veterinary medical science, animal science, and related fields
G. Biology	43. Biology at molecular to cellular levels, and related fields
	44. Biology at cellular to organismal levels, and related fields
	45. Biology at organismal to population levels, and anthropology, and related fields
	46. Neuroscience and related fields
H. Pharmacy	47. Pharmaceutical sciences and related fields
	48. Biomedical structure and function and related fields
	49. Pathology, infection/immunology, and related fields
I. Medicine	50. Oncology and related fields
	51. Brain sciences and related fields
	52. General internal medicine and related fields

Broad Section	Medium-sized Section
	53. Organ-based internal medicine and related fields
	54. Internal medicine of the bio-information integration and related fields
	55. Surgery of the organs maintaining, homeostasis and related fields
	56. Surgery related to the biological and sensory functions and related fields
	57. Oral science and related fields
	58. Society medicine, nursing, and related fields
	59. Sports sciences, physical education, health sciences, and related fields
J. Informatics	60. Biomedical engineering and related fields
	61. Human informatics and related fields
	62. Applied informatics and related fields Basic Section
K. Environment Science	63. Environmental analyses and evaluation and related fields
	64. Environmental conservation measure and related fields

Source: JSPS, Grants-in-Aid for Scientific Research, https://www.jsps.go.jp/file/storage/kaken_kiban_2024_g_2307/review_section_table_e.pdf, accessed 17/10/2024

2.3 International Organization: OECD cases

The OECD introduced the Fields of Science and Technology (FOS) classification system in 2002 to standardize the statistical categorization of academic and technical disciplines. It comprises six key categories with 43 sub-categories as shown in [Table OO]. This system was developed to facilitate the exchange of data related to research facilities and outcomes. In 2007, it was updated and renamed the Revised Fields of Science and Technology.

[Table 5-3] OECD Field of Science and Technology (revision in 2007)

Disciplines		Knowledge Fields
1	NATURAL SCIENCES	1.1 MATHEMATICS 1.2 COMPUTER AND INFORMATION SCIENCES 1.3 PHYSICAL SCIENCES 1.4 CHEMICAL SCIENCES 1.5 EARTH AND RELATED ENVIRONMENTAL SCIENCES 1.6 BIOLOGICAL SCIENCES 1.7 OTHER NATURAL SCIENCES
2	ENGINEERING AND TECHNOLOGY	2.1 CIVIL ENGINEERING 2.2 ELECTRICAL ENGINEERING, ELECTRONIC ENGINEERING, INFORMATION ENGINEERING 2.3 MECHANICAL ENGINEERING 2.4 CHEMICAL ENGINEERING 2.5 MATERIALS ENGINEERING 2.6 MEDICAL ENGINEERING 2.7 ENVIRONMENTAL ENGINEERING 2.8 ENVIRONMENTAL BIOTECHNOLOGY 2.9 INDUSTRIAL BIOTECHNOLOGY 2.10 NANO-TECHNOLOGY 2.11 OTHER ENGINEERING AND TECHNOLOGIES
3	MEDICAL AND HEALTH SCIENCES	3.1 BASIC MEDICINE 3.2 CLINICAL MEDICINE 3.3 HEALTH SCIENCES 3.4 HEALTH BIOTECHNOLOGY 3.5 MATERIALS ENGINEERING
4	AGRICULTURAL SCIENCES	4.1 AGRICULTURE, FORESTRY, AND FISHERIES 4.2 ANIMAL AND DAIRY SCIENCE 4.3 VETERINARY SCIENCE 4.4 AGRICULTURAL BIOTECHNOLOGY 4.5 OTHER AGRICULTURAL SCIENCES
5	SOCIAL SCIENCES	5.1 PSYCHOLOGY 5.2 ECONOMICS AND BUSINESS 5.3 EDUCATIONAL SCIENCES 5.4 SOCIOLOGY 5.5 LAW 5.6 POLITICAL SCIENCE 5.7 SOCIAL AND ECONOMIC GEOGRAPHY 5.8 MEDIA AND COMMUNICATIONS 5.9 OTHER SOCIAL SCIENCES

Disciplines		Knowledge Fields
6	HUMANITIES	6.1 HISTORY AND ARCHAEOLOGY 6.2 LANGUAGES AND LITERATURE 6.3 PHILOSOPHY, ETHICS AND RELIGION 6.4 ART (ARTS, HISTORY OF ARTS, PERFORMING ARTS, MUSIC) 6.5 OTHER HUMANITIES

Source: OECD, Revised Field of Science and Technology Classification in the Frascati Manual, https://www.britishcouncil.cl/sites/default/files/oecd_disciplines_british_council.pdf, accessed 16/10/2024

Galindo-Rueda and Verger (2016) from OECD also proposed a novel classification system for industries based on their R&D intensity, defined as the ratio of R&D expenditure to value added in each sector. They divided industries into five categories: high, medium-high, medium, medium-low, and low R&D intensity, applicable to both manufacturing and non-manufacturing sectors. This approach links science and technology fields with R&D intensity according to the International Standard Industrial Classification (ISIC). This taxonomy effectively groups industries by their R&D intensity, suggesting the potential for developing new classification systems tailored to various objectives.

[Table 5-4] OECD Taxonomy of Economic Activities based on R&D Intensity

	Manufacturing	R&D as % of GVA[2]	Non-manufacturing	R&D as % of GVA[2]
High R&D intensity industries	303[1] : Air and spacecraft and related machinery	31.69	72 : Scientific research and development	30.39
	21 : Pharmaceuticals	27.98	582[1] : Software publishing	28.94
	26 : Computer, electronic and optical products	24.05		
Medium-high R&D intensity industries	252[1] : Weapons and ammunition	18.87	62-63: IT and other information services	5.92
	29 : Motor vehicles, trailers and semi-trailers	15.36		
	325[1] : Medical and dental instruments	9.29		
	28 : Machinery and equipment n.e.c.	7.89		
	20 : Chemicals and chemical products	6.52		
	27 : Electrical equipment	6.22		
	30X[1] : Railroad, military vehicles and transport n.e.c. (ISIC 302, 304 and 309)	5.72		
Medium R&D intensity industries	22 : Rubber and plastic products	3.58		
	301[1] : Building of ships and boats	2.99		
	32X[1] : Other manufacturing except medical and dental instruments (ISIC 32 less 325)	2.85		
	23 : Other non-metallic mineral products	2.24		
	24 : Basic metals	2.07		
	33 : Repair and installation of machinery and equipment	1.93		
Medium-low R&D intensity industries	13 : Textiles	1.73	69-75X: Professional, scientific and technical activities except scientific R&D (ISIC 69 to 75 less 72)	1.76
	15 : Leather and related products	1.65	61 : Telecommunications	1.45
	17 : Paper and paper products	1.58	05-09: Mining and quarrying	0.80
	10-12: food products, beverages and tobacco	1.44	581[1] : Publishing of books and periodicals	0.57
	14 : Wearing apparel	1.40		
	25X[1] : Fabricated metal products except weapons and	1.19		

	Manufacturing	R&D as % of GVA[2]	Non-manufacturing	R&D as % of GVA[2]
	ammunition (ISIC 25 less 252)			
	19 : coke and refined petroleum products	1.17		
	31 : Furniture	1.17		
	16 : Wood and products of wood and cork	0.70		
	18 : Printing and reproduction of recorded media	0.67		
Low R&D intensity industries			64-66: Financial and insurance activities	0.38
			35-39: electricity, gas and water supply, waste management and remediation	0.35
			59-60: Audiovisual and broadcasting activities	0.32
			45-47: Wholesale and retail trade	0.28
			01-03: Agriculture, forestry and fishing	0.27
			41-43: construction	0.21
			77-82: Administrative and support service activities	0.18
			90-99: Arts, entertainment, repair of household goods and other services	0.11
			49-53: Transportation and storage	0.08
			55-56: Accommodation and food services activities	0.02
			68 : Real estate activities	0.01

1. Higher-level industries of the ISIC hierarchy are also classified and the classification is performed at the 2-digit level.
2. The classification is based on aggregated R&D intensities. Value added and R&D of the 29 economies of the sample are aggregated using purchasing power parities.

Source: extracted from Galindo-Rueda, F. and F. Verger (2016, p.10)

3. Implications towards Cambodian System

As discussed earlier, the science and technology standard classification system serves as a crucial tool for many policy purposes. The system can provide valuable directions and guidelines that can foster private sector growth and international collaboration. It also offers an objective basis for governments to set R&D investment priorities, aiding in the rational formulation of science and technology policies. It can be also a good measure for strategic technology selection and priority analysis in the development of future industries. By utilizing this system, key technology fields and cooperative technology areas can be clearly identified, and the progress of specific or emerging technology fields can be effectively monitored, which is one of most important policy issues. Additionally, it helps prevent redundant investments and resource wastage in R&D by supporting focused investment in strategic technologies, thereby ensuring efficient resource allocation.

For developing countries, it is essential to understand the utility of the science and technology classification systems of advanced nations while also creating adaptive systems tailored to their unique industrial and economic contexts as shown in the cases of the Republic of Korea and OECD. Relying solely on the systems of advanced countries without considering local technological priorities and characteristics can lead to failure to meet domestic industrial needs. Therefore, developing an adaptive classification system that reflects the specific industrial conditions of the country is necessary. In developing countries, collaboration between industry and academia is crucial. The industry sector may not require advanced science and technology for cutting-edge knowledge and high-end products as much. Instead, there is a greater need for industrial technologies that enhance the quality of existing products and processes. Therefore, the classification system should be tailored to meet these specific needs.

CHAPTER 6

Conclusion

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Chapter 6. Conclusion

1. Outcomes and Expected Benefits

The first output of the 2024 Program can be found from the Tech Show and Kick-Off Workshop during the Cambodia's National STI Day. In March 2024, the workshop included a comprehensive presentation of the findings from the National R&D Management System demand survey, followed by a detailed demonstration of the NTIS platform. These activities were complemented by in-depth discussions on the system's relevance to Cambodia's R&D landscape and its potential applications in addressing the country's pressing R&D needs effectively.

The second output of the 2024 program is the concept design of Cambodia's national R&D information system with key functions, data portfolio to be collected and monitored, and estimated budget cases based on two scenarios. MISTI and STEPI, with the technical support from KISTI, which has the know-how and experiences in building a national R&D information management system and NTIS, established a target system design to build a Cambodia's National R&D Information Management System/Platform (CTIS) that is in line with all relevant Cambodia's policies and utilizes the accumulated information.

Based on the concept design produced, STEPI held a stakeholder workshop, as a side event of ASEAN-COSTI 85 on June 6 2024, in Siem Reap Cambodia. The event featured the presentations of the initial system design draft and the collection of feedback from key stakeholders. The first session contains background and needs assessment results for the Cambodia's National R&D Information Management System (CTIS). The second session topic was a concept and development plan for the CTIS. During the panel discussion, timelines of implementation, anticipated benefits and R&D budget were discussed.

The fourth output of the 2024 program is the study on Cambodia's Legal and Regulatory Supporting Framework on R&D. In order to do so, a service contract was officially signed between STEPI and MISTI. The purpose of this contract is aimed to serve as a way to assess the feasibility of developing a CTIS. This assessment included a detailed review of the relevant legal and institutional frameworks supporting research and development (R&D) in Cambodia, in national core and sectoral policy perspectives.

Lastly, the utility of science and technology standard classification system was discussed by the analysis on the cases of advanced East Asian countries: Korea, China, and Japan. It helps governments to systematically classifying and organizing the nation's science and technology activities through the stand classification system, including R&D programs and projects, R&D equipment, and R&D outputs of journal papers and patents. For the effective operation of a national R&D information management system, these information and activities should be classified and organized systematically so that it can be used as a basic tool for policy makers to effectively manage and evaluate research and development (R&D) activities, technological innovation, and related policies. Therefore, the classification system for a country can be used as a guideline for establishing a country's science and technology policies including R&D monitoring, R&D investment decisions, and performance evaluation, a guideline for efficiently distributing and managing national R&D resources, and to compare the scientific and technological levels of a country and the comparable countries in international scientific and technological cooperation and to explore opportunities for international cooperation based on this.

2. Policy Implications and Future Plan

The concept design of Cambodia's national R&D information system (CTIS) paves a new way for the Research and Development (R&D) landscape in Cambodia and offers an overview of the transformative future for the landscape of scientific direction that is envisioned by the research community in Cambodia. The consensus building process among the key stakeholders enabled MISTI and STEPI to reflect collective insights on the design process of an ideal R&D information management system that can effectively address the evolving needs and tackle the challenges of researchers in the country. The emphasis on features such as a robust research database, including researchers, project proposals, funding resources, publications demonstrates a clear vision for a robust R&D information ecosystem that caters to the needs of diverse researchers and scientists. This ideal system serves as a catalyst for creativity, encouraging interdisciplinary collaboration, increasing transparency, and promoting effective resource utilization in the Cambodian government.

As we navigate the future of the research landscape in Cambodia, it is clear that the ideal R&D information and management system, as envisioned by the research community, will play a critical role in determining how information is generated, shared, and implemented within MISTI and Cambodia's research community. The preliminary feasibility study provides useful information for stakeholders and developers to align their efforts with the real needs and goals of CTIS in the Cambodian STI context. To achieve the optimal result and the widespread use of the system and long-term sustainability, the proposed system must be adaptable to changing technologies, support diverse research methodologies, and, most importantly, be user-friendly.

The findings of this report provide a concrete roadmap for developing and implementing an R&D information and management system that is aligned with the dynamic and collaborative framework of the research management that reflects the needs of the researchers in Cambodia. The shared vision of the research community in Cambodia provides a basis for fostering innovation, promoting technology transfer, leveraging scientific progress, and contributing to the ultimate goal for socio-economic development in Cambodia. In this regard, MISTI made huge efforts to convene key stakeholders across ministries, academia and private sector actors, share the information on CTIS and persuade them to be included in the information loop.

Based on the results of the preliminary feasibility study on the establishment of Cambodia national R&D information system (CTIS), STEPI research team plans to provide consulting on designing the implementation of Cambodia's national R&D information system in 2025. In 2025, STEPI will also support MISTI to develop a more detailed operational plan for its national R&D information system in order for MISTI to engage national key stakeholders to collaborate with for the smooth operation of CTIS. Korean experts from KISTI and R&D management related public and private experts will be also invited to participate to meet the goal of the programs. In addition, in cooperation with MISTI, STEPI team plans to support in capacity enhancement of MISTI officials who will be responsible for the implementation plan and operation of the national R&D information system for the Cambodian government that will improve the governance and ecosystem of R&D management in Cambodia.

REFERENCES

CHATPER 2. Main Activities

Royal Government of Cambodia (2023). Pentagonal Strategy-Phase I.

Royal Government of Cambodia (2019). National Strategic Development Plan 2019-2023.

Royal Government of Cambodia (2018). Cambodian Sustainable Development Goals (CSDGs) Framework 2016-2030.

Royal Government of Cambodia (2019). The National STI Policy 2020-2030.

Ministry of Industry, Science, Technology and Innovation (2023). National Research Agenda 2025.

Ministry of Industry, Science, Technology and Innovation (2021). Cambodia's STI Roadmap 2030.

Ministry of Industry, Science, Technology and Innovation (2023). Energy Technology Roadmap.

Ministry of Industry, Science, Technology and Innovation (2023). Tourism Technology Roadmap.

Ministry of Industry, Science, Technology and Innovation (2023). Digital Technology Roadmap.

Ministry of Industry, Science, Technology and Innovation (2022). Agriculture Technology Roadmap.

Ministry of Industry, Science, Technology and Innovation (2022). Health Technology Roadmap.

Ministry of Industry, Science, Technology and Innovation (2022). Education Technology Roadmap.

Ministry of Education Youth and Sports (2014). Policy on Higher Education Vision 2030.

Ministry of Education Youth and Sports (2011). Master Plan for Research Development in the Education Sector 2011-2015.

Supreme National Economic Council (2021). Digital Economy and Society Policy Framework 2021-2035.

Ministry of Posts and Telecommunication (2014). Cambodian Information and Communication Technology Masterplan 2020.

Ministry of Environment (2021). Cambodia's Circular Economy Strategy and Action Plan.

Ministry of Environment (2013). Cambodia's Climate Change Strategic Plan 2014- 2023.

Ministry of Agriculture, Forestry and Fisheries (2019). Agricultural Sector Strategic Development Plan 2019-2023.

Ministry of Health (2016). Health Strategic Plan 2016-2020.

Council for the Development of Cambodia (2015). Industrial Development Policy and Plan 2015-2025.

Council for the Development of Cambodia (2015). Law on Investment.

CHAPTER 3. Cambodia's Legal and Regulatory Supporting Framework

KISTI (2016), "Development on Platform Model for National R&D Information of Costa Rica."

KISTI NTIS Center, 「NTIS Introduction」.

KISTI NTIS Center, 「Vietnam FS study」.

Korea Software Industry Association, 「'24 A guide to calculating the price of SW business」.

National Science and Technology Commission (2004), 「NTIS Construction Plan」.

NTIS (National Science & Technology Information Service), www.ntis.go.kr

CHAPTER 4. Cambodia's National R&D Information Management System (CTIS) Concept and Development Plan of Actions

Lee, S., Park, C., and J. Moon (2023), Improvement and Application of the National Science and Technology Standard Classification System, KISTEP 2023-005, https://www.kistep.re.kr/reportDetail.es?mid=a10305020000&rpt_tp=831-006&rpt_no=RES0220230157, accessed 16/10/2024. (in Korean)

Galindo-Rueda, F. and F. Verger (2016), "OECD Taxonomy of Economic Activities Based on R&D Intensity", OECD Science, Technology and Industry Working Papers, No.2016/04, OECD Publishing, Paris, https://doi.org/10.1787/5jlv73sqqp8r-en, accessed 16/10/2024.

Japan Society for Promotion of Science, Grants-in-Aid for Scientific Research, https://www.jsps.go.jp/file/storage/kaken_kiban_2024_g_2307/review_section_table_e.pdf, accessed 17/10/2024